

Sickle Cell Outcome Grading System





Forward

Sickle Cell Outcome Grading System (SCOGS) is a new method for classifying disease severity for sickle cell disease (SCD).

SCOGS was funded through the National Institutes of Health (NIH) Research Project grant #5R01CA270157 and developed by adapting the model of Common Terminology Criteria for Adverse Events (CTCAE) severity grading index to fit within the context of SCD.

This system covers all major disease complications and includes definitions, diagnostic criteria, severity gradings (1-5), and frequency classifications (acute, chronic, or chronic with exacerbation) for 53 health outcomes.

To achieve standardized and widely applicable uses, SCOGS utilizes objective measures as much as possible, avoids dependencies on treatment, and follows a standard framework. Any deviations from the framework are noted within the individual outcome classification tables.

Framework

To best utilize severity grades, each health outcome should be graded and counted by event, with consideration to the frequency of the complication. Guidance for these frequency patterns are denoted in SCOGS health outcome classification tables as acute, chronic, or chronic with exacerbation.

- **Acute** – Sudden onset of symptoms that are typically severe and resolve. Their duration may vary, and most have a short duration (i.e., < 1 week in splenic sequestration)
- **Chronic** – A condition that usually develops gradually, persists over time, and may or may not resolve
- **Chronic with Exacerbations** – A chronic condition that worsens or flares up, leading to an acute worsening of symptoms

Reference the [Grading Frequency Guide](#) for a full table of health outcomes by frequency category.

Grade 1

Evidence of specific health outcome event that is usually mild. Whenever objective measures of disease severity exist (e.g., biomarkers, established severity grading), these are used with or without health utilization data.

Otherwise, events that may be treated at home, or if they require a visit to a medical facility for treatment, they will not require admission, are classified as Grade 1.

Grade 2

A specific health outcome event of greater severity than Grade 1. Symptoms are usually mild to moderate. Whenever objective measures of disease severity exist (e.g., biomarkers, established severity grading), these are used with or without health utilization data.

Otherwise, events that cannot be treated at home and require a visit to a medical facility for treatment but do not require an admission, and symptoms are objectively more severe than those in Grade 1 are classified as Grade 2.

Grade 3

A specific health outcome event of greater severity than Grade 2. Symptoms are severe, and medical treatment is used to treat the condition/complication. Whenever objective measures of disease severity exist (e.g., biomarkers, established severity grading), these are used with or without health utilization data.

Otherwise, events that require a visit to a medical facility for treatment of severe complications and may require admission (usually do not require critical care) and are objectively more severe than those in Grade 2 are classified as Grade 3.

Grade 4

A specific health outcome event of greater severity than Grade 3. Symptoms are severe or life-threatening, and medical treatment is used to treat the condition/complication. Whenever objective measures of disease severity exist (e.g., biomarkers, established severity grading), these are used with or without health utilization data.

Otherwise, events that require a visit to a medical facility for treatment of severe complications and may require admission (usually they require critical care) and are objectively more severe than those in Grade 3 are classified as Grade 4.

Grade 5

Death related to the specific health outcome.

Development

The methodological development of SCOGS included 3 rounds of Delphi consensus building and validation in addition to consultations with disease area and specialty experts. The Delphi panelists included 28 hematologists and 1 pulmonologist, with clinical and research practices serving both pediatric and adult SCD populations, from regions across the United States. 21 of these panelists also participated during an in-person workshop to finalize edits to the 53 health outcomes, each of which reached our consensus criteria of > 70% agreement with the proposed classification.

Applications

This formalized grading system has widespread implications and applications across a spectrum of providers, patients, and advocacy realms. SCOGS should be used as a multi-faceted tool for clinical decision-making, in clinical trial design for inclusion criteria and endpoints, to standardize assessment of disease modifying therapy and treatment response, and to inform various methods for predictive risk modeling.





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Health Outcomes by Organ/System

Cardiovascular

Arrhythmia

Chronic with Exacerbation

Grade 1	Asymptomatic AND NO intervention indicated
Grade 2	Symptomatic AND non urgent medical intervention indicated
Grade 3	Symptomatic AND urgent intervention indicated (ablation, etc)
Grade 4	Arrhythmia complications requiring cardioversion AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments. ²
Grade 5	Death from arrhythmia

Definition

A condition characterized by abnormal heart rhythm.¹

Diagnostic Criteria

Abnormal atrial and/or ventricular rhythm as identified by electrocardiogram.

Methodology

Modified CTCAE v5 grading system for “Ventricular arrhythmia”³ to include specific interventions and level of care required.

References

1. What is an Arrhythmia? American Heart Association. Accessed April 25, 2024. <https://www.heart.org/-/media/files/health-topics/answers-by-heart/what-is-arrhythmia.pdf>
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. *Ann Intensive Care*. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5
3. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Deep Vein Thrombosis (DVT)

Acute

Grade 1	Not requiring intervention
Grade 2	Requiring medical intervention in the form of anticoagulation AND NOT requiring invasive intervention
Grade 3	Requiring urgent invasive medical intervention in the form of thrombolysis (site directed TPA) AND NO surgical thrombectomy
Grade 4	Limb or life-threatening consequences OR hemodynamic or neurologic instability with or without surgical thrombectomy
Grade 5	Death in the setting of DVT

Definition

Deep vein thrombosis (DVT) refers to the formation of one or more blood clots (a blood clot is also known as a “thrombus,” while multiple clots are called “thrombi”) in one of the body’s largest veins, most commonly the lower limbs (e.g., lower leg or calf).¹

Diagnostic Criteria

Presence of deep venous occlusion on imaging study, including ultrasound, CT, MRI, venogram, excluding pulmonary embolism. Veins may include, but not limited to, common femoral vein, greater saphenous vein, superficial femoral vein, popliteal vein, peroneal vein, jugular vein, brachiocephalic vein, subclavian vein, and axillary vein.

Methodology

Grading system based on higher levels of care and interventions required.

References

1. (US) OotSG, National Heart L, and Blood Institute (US). (US) OotSG, ed. The Surgeon General’s Call to Action to Prevent Deep Vein Thrombosis and Pulmonary Embolism. 2008. www.ncbi.nlm.nih.gov/books/NBK44184/

Diastolic Dysfunction

Chronic

Grade 1	N/A
Grade 2	Diastolic Dysfunction Present
Grade 3	N/A
Grade 4	N/A
Grade 5	N/A

Definition

Disturbance in ventricular relaxation, distensibility or filling—where the ejection fraction (EF) is normal and whether the patient is asymptomatic or symptomatic.¹

Diagnostic Criteria

Adults ≥ 18 years of age. According to the 2016 American Society of Echocardiography (ASE) guidelines the following four 2D-echocardiogram variables should be used to diagnose for abnormal diastolic function in the adult general population with normal left ventricular systolic function.²

Thresholds have been defined above or below, which are considered abnormal.

LV diastolic function is abnormal **only if two or more of these variables are outside the normative range²:**

- Average $E/e' > 14$ (lateral $E/e' > 13$ or septal $E/e' > 15$)
- Annular e' velocity: septal $e' < 7\text{cm/sec}$ or lateral $e' < 10\text{cm/sec}$
- $\text{TRV} > 2.8\text{m/sec}$
- Left atrial volume index $> 34\text{ mL/m}^2$

Diastolic Dysfunction

Chronic

Methodology

The grading system was created using ASE guidelines to accommodate SCD conditions. Grading is simplified due to a lack of consensus on the classification of diastolic dysfunction in SCD. Outcome grading was created with the assistance of expert adult cardiologist consultants.

*This is an exploratory outcome that will require further data collection and validation. It did not undergo all Delphi rounds for approval.

*Current definition and grading system may be updated following ASE 2025 guidelines.

References:

1. Leite-Moreira AF. Current perspectives in diastolic dysfunction and diastolic heart failure. *Heart*. 2006;92(5):712-718. doi:10.1136/hrt.2005.062950
2. Nagueh SF, Smiseth OA, Appleton CP, et al. Recommendations for the Evaluation of Left Ventricular Diastolic Function by Echocardiography: An Update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. Apr 2016;29(4):277-314. doi:10.1016/j.echo.2016.01.011

Heart Failure Exacerbation

Acute

Grade 1	N/A
Grade 2	Heart failure exacerbation AND NOT requiring intervention
Grade 3	Heart failure exacerbation AND requiring intervention in the form of diuretics and/or nitric oxide containing compounds and/or non-invasive ventilation AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ⁵
Grade 4	Heart failure exacerbation AND requiring intervention in the form of diuretics and/or nitric oxide containing compounds and/or non-invasive ventilation AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ⁵
Grade 5	Death from heart failure exacerbation

Definition

Acute heart failure (or heart failure exacerbation) is defined as gradual or rapid change in heart failure signs and symptoms resulting in a need for urgent therapy. These symptoms are primarily the result of severe pulmonary congestion due to elevated left ventricular filling pressures. Acute heart failure can occur in patients with preserved or reduced ejection fraction. Concurrent cardiovascular conditions such as coronary heart disease, hypertension, valvular heart disease, atrial arrhythmias, and/or non-cardiac conditions (such as renal dysfunction, diabetes, and anemia) are often present and may precipitate or contribute to the pathophysiology of the exacerbation.¹

Heart Failure Exacerbation

Acute

Diagnostic Criteria

Workup and diagnosis of heart failure exacerbation include a combination of physical exam and laboratory findings and should include some or all of the following²:

- Elevated jugular venous pressure (JVP) on exam
- Signs of peripheral edema
- Orthopnea or dyspnea
- Electrocardiogram (EKG) with or without atrial arrhythmia
- Chest X-ray (CXR) showing signs of cardiac enlargement and/or pulmonary congestion
- Elevated (NT-pro) brain natriuretic peptide (NT-pro BNP or BNP)
- Elevated creatinine and/or blood urea nitrogen (BUN)
- Low serum sodium
- Elevated troponin

Methodology

Grading system was created using AHA/ACC/HFSA guidelines⁴ to accommodate SCD conditions, treatment, and interventions. Grading was informed by expert adult pulmonology consultation.

References

1. Gheorghiade M, et al. Acute Heart Failure Syndromes. *Circulation*. 2005;112(25):10.
2. Ural D, Çavuşoğlu Y, Eren M, et al. Diagnosis and management of acute heart failure. *Anatol J Cardiol*. Nov 2015;15(11):860-89. doi:10.5152/AnatolJCardiol.2015.6567
3. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. *Ann Intensive Care*. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5
4. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. May 3 2022;145(18):e895-e1032. doi:10.1161/cir.0000000000001063

Myocardial Infarction

Acute

Grade 1	N/A
Grade 2	Abnormal cardiac enzymes as listed above AND NO severity criteria, essentially non-ST elevation myocardial infarction (NSTEMI)
Grade 3	Myocardial infarction with ANY of the severity criteria AND NO hemodynamic instability
Grade 4	Myocardial infarction resulting in significant cardiovascular compromise AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ³ (with or without coronary intervention)
Grade 5	Death from myocardial infarction

Definition

Myocardial cell death (necrosis) due to prolonged ischemia.¹

Diagnostic Criteria

Diagnosed by one of three methods.

1. Detection of serum cardiac biomarkers:

- MB-CK > 9 IU/L at 6 to 12 hours from the onset of symptoms; peak from 14 to 36 hours
- Troponin-T > 0.1 ng/mL at 6 to 12 hours from the onset of symptoms; peak from 24 to 36 hours

AND at least one of the following:

- ECG changes indicative of new ischemia (new ST-T changes or new left bundle branch block [LBBB])

Myocardial Infarction

Acute

Diagnostic Criteria (*continued*)

- ECG manifestations of acute myocardial ischemia (in the absence of left ventricular hypertrophy and LBBB)
 - ST elevation: new ST elevation at the J-point in two contiguous leads with the cut-off points of ≥ 0.2 mV in men or ≥ 0.15 mV in women in leads V2–V3 and/or ≥ 0.1 mV in other leads
 - ST depression and T-wave changes: new horizontal or down-sloping ST depression ≥ 0.05 mV in two contiguous leads; and/or T inversion ≥ 0.1 mV in two contiguous leads with prominent R-wave or R/S ratio ≥ 1
 - Development of pathological Q-waves
 - Imaging evidence of new loss of viable myocardium or new regional wall abnormality **OR**
2. Sudden, unexpected cardiac death, involving cardiac arrest, often with symptoms suggestive of myocardial ischemia **OR**
 3. Histology/pathology (postmortem, most definitive):
 - Negative staining of the infarcted area with nitro blue tetrazolium is evident at 6 to 8 hours
 - Hematoxylin-fuchsin-picric acid stain is independent of autolysis and is positive for infarct at 30 minutes (useful in sudden death)¹

Severity Criteria:

Based on Atherosclerosis Risk in Communities (ARIC) study²

- SBP < 100 mm Hg
- Abnormal pulse rate
- Diagnostic ECG with ST elevation
- New Q wave on ECG
- New left bundle branch block

Myocardial Infarction

Acute

Methodology

Grading system was based on higher levels of care and interventions required. Outcome grading was created with assistance of expert adult cardiologist consultant.

References

1. Ballas SK, Lief S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Myerson M, Coady S, Taylor H, Rosamond WD, Goff DC. Declining Severity of Myocardial Infarction From 1987 to 2002. Circulation. 2009;119(4):503-514. doi:10.1161/CIRCULATIONAHA.107.693879
3. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5

Systemic Arterial Hypertension

Chronic

Grade 1	No medical intervention required (no use of medication for hypertension) AND • Ages 1-13 years: BP $\geq$ 90th percentile to $<$ 95th percentile for age; 120/80 mmHg to $<$ 95th percentile (whichever is lower) • Ages $>$ 13 years: BP 120/$<$ 80 to 129/$<$ 80 mmHg
Grade 2	Stage 1 Hypertension. Requiring no more than one anti-hypertensive drug OR • Ages 1-13 years: BP $\geq$ 95th percentile to $<$ 95th percentile + 12 mmHg; BP range within 130/80 to 139/89 mmHg (whichever is lower) • Ages $>$ 13 years: BP 130/80 to 139/89 mmHg
Grade 3	Stage 2 Hypertension. Requiring two or more anti-hypertensives OR • Ages 1-13 years: BP $\geq$ 95th percentile + 12 mmHg, or $\geq$ 140/90 mmHg (whichever is lower) • Ages $>$ 13 years: $\geq$ 95th percentile + 12 mmHg; $\geq$ 140/90 mmHg (whichever is lower)
Grade 4	Elevated BP resulting in end organ damage (e.g., acute kidney injury, sudden vision loss, stroke, myocardial infarction, pulmonary edema) at any age
Grade 5	Death from malignant hypertension

Systemic Arterial Hypertension

Chronic

Definition

Blood pressure exceeding the 90th percentile for age.¹

Diagnostic Criteria

Evaluation focuses on determining possible causes of and/or comorbidities associated with hypertension. Evaluation should include appropriate patient history, family history, physical examination, laboratory evaluation, and imaging.² Diagnosis should be made based on baseline blood pressure and exclude times when patient is experiencing an acute sickle cell pain event.

Methodology

Grading system combines guidelines set forth by the American College of Cardiology (ACC) and American Heart Association (AHA) with separate guidelines for adults³ and pediatrics², and includes medical interventions or sequela. ACC/AHA Stages 1 and 2 modified to grades 2 and 3, respectively.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. *Am J Hematol.* Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Flynn JT, Kaelber DC, Baker-Smith CM, et al. Clinical Practice Guideline for Screening and Management of High Blood Pressure in Children and Adolescents. *Pediatrics.* 2017;140(3)doi:10.1542/peds.2017-1904
3. Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension.* 2018;71(6):e13-e115. doi:doi:10.1161/HYP.0000000000000065

Systolic Dysfunction

Chronic

Grade 1	N/A
Grade 2	Mild dysfunction (LVEF 40-49%) with or without the need for medical or surgical intervention
Grade 3	Moderate dysfunction (LVEF 30-39%) with or without the need for medical or surgical intervention
Grade 4	Severe dysfunction (LVEF < 30%) with or without the need for medical or surgical intervention
Grade 5	N/A

Definition

Complex clinical syndrome with symptoms and signs that result from any structural or functional impairment of left ventricular ejection fraction (LVEF) of blood.¹

Diagnostic Criteria

LVEF can be calculated using various modalities, either subjectively by using visual estimation or objectively by quantitative methods. The preference is to employ quantitative measures to assess LVEF to minimize variability and favor more precision and accuracy in measurement. The following modalities may be used²:

- Echocardiography (ECHO)
- Magnetic resonance imaging (MRI)
- Computed tomography (CT)
- Gated equilibrium radionuclide angiography (MUGA)
- Gated myocardial perfusion imaging with either single-photon emission computed tomography (SPECT) or positron emission tomography (PET)
- Left ventricular contrast ventriculography during invasive catheterization

Systolic Dysfunction

Chronic

Methodology

Grading system created using AHA/ACC/HFSA guidelines¹ to accommodate SCD conditions, treatment, and interventions. Grading informed by expert adult pulmonology consultation.

References

1. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. May 3 2022;145(18):e895-e1032. doi:10.1161/cir.0000000000001063
2. Kosaraju A, Goyal A, Grigorova Y, Makaryus AN. Left Ventricular Ejection Fraction. StatPearls. StatPearls Publishing; Updated 2023 April 24. www.ncbi.nlm.nih.gov/books/NBK459131/

Tricuspid Regurgitant Jet Velocity (TRV) Elevation on Echocardiogram

Chronic

Grade 1	N/A
Grade 2	N/A
Grade 3	TRV 2.5-2.9 m/second
Grade 4	TRV $\geq$ 3.0 m/second
Grade 5	N/A

Definition

The Tricuspid Regurgitant jet velocity (TRV) is an index of the systolic pressure gradient between the right ventricle and the right atrium.¹

In adult and pediatric SCA studies, TRV $\geq$ 2.5m/sec is considered elevated.

Diagnostic Criteria

TRV is measured as per echocardiogram at steady state and not during hospitalization or within 2 weeks of vaso-occlusive event.

Methodology

Criteria based on TRV index readings of “Mild-Moderate and Severe” in association with pulmonary hypertension within SCD literature.²

References

1. Jang AY, Shin MS. Echocardiographic Screening Methods for Pulmonary Hypertension: A Practical Review. J Cardiovasc Imaging. Jan 2020;28(1):1-9. doi:10.4250/jcvi.2019.0104
2. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550



Health Outcomes by Organ/System

Central Nervous System

—

Cerebral Vasculopathy

Chronic

Grade 1	Mild stenosis (25-49%) of ≤ 2 arterial segments as per MRA
Grade 2	Mild stenosis (25-49%) of > 2 arterial segments as per MRA
Grade 3	Moderate stenosis (50-74%) of ≤ 2 arterial segments as per MRA
Grade 4	Moderate stenosis (50-74%) of > 2 arterial segments OR Severe stenosis/occlusion ($\geq 75\%$) of ≤ 2 arterial segments as per MRA
Grade 5	Severe stenosis/ occlusion ($\geq 75\%$) of ≥ 3 arterial segments as per MRA OR presence of moyamoya syndrome

Definition

Stenosis or abnormal vasculature caused by elevated erythrocyte sickling in intracerebral arteries that places patients with sickle cell disease at higher risk for stroke or transient ischemic infarct.¹

Diagnostic Criteria

- Conventional angiography or Brain MRA examinations are used to evaluate vessel stenoses in four segments of each internal carotid artery (ICA), and 3 segments of each anterior cerebral artery (ACA) and middle cerebral artery (MCA).
- Vessel stenosis is recorded by vascular location, frequency (number of stenoses), and severity (percentage vessel occlusion).
- Severity of vessel segment occlusion is graded as mild (25-49%), moderate (50-74%), severe (75-99%), or occlusion ($> 99\%$).¹

Cerebral Vasculopathy

Chronic

Methodology

Severity is based on magnetic resonance imaging or angiography results. Severity categories of mild stenosis, moderate stenosis, and severe stenosis/occlusion exist, and we adapted the categories to fit 5 grades. Grades 5 and 6 in Helton's 6-point grading scale¹ were condensed into Grade 5 in our grading system, which includes moyamoya syndrome.²

References

1. Helton KJ, Adams RJ, Kesler KL, et al. Magnetic resonance imaging/angiography and transcranial Doppler velocities in sickle cell anemia: results from the SWiTCH trial. *Blood*. Aug 7 2014;124(6):891-8. doi:10.1182/blood-2013-12-545186
2. Demartini Z, Jr., Teixeira BC, Koppe GL, Gatto LAM, Roman A, Munhoz RP. Moyamoya disease and syndrome: a review. *Radiol Bras*. Jan-Feb 2022;55(1):31-37. doi:10.1590/0100-3984.2021.0010

Chronic Pain

Chronic

Grade 1	No unplanned visits to medical facilities for pain treatment within a 12-month period OR at least one of the below in the past 3 months: • Pain and Hurt Score ≥ 81 • PROMIS T-score ≤ 48 on pain interference • PROMIS T-score ≤ 41 on pain behavior
Grade 2	1-3 unplanned visits to medical facilities for pain treatment within a 12-month period OR at least one of the below in the past 3 months: • Pain and Hurt Score 61-80 • PROMIS T-score > 48 but ≤ 64 on pain interference • PROMIS T-scores > 41 but ≤ 57 on pain behavior
Grade 3	4-10 unplanned visits to medical facilities for pain treatment within a 12-month period OR at least one of the below in the past 3 months: • Pain and Hurt Score ≤ 60 • PROMIS T-score > 64 on pain interference • PROMIS T-score > 57 on pain behavior
Grade 4	> 10 unplanned visits to medical facilities for pain treatment within a 12-month period OR at least two of the below in the past 3 months: • Pain and Hurt Score ≤ 60 • PROMIS T-score > 64 on pain interference • PROMIS T-score > 57 on pain behavior
Grade 5	N/A

Chronic Pain

Chronic

Definition

SCD pain that is present on most days and has occurred for at least the previous 3 months (this is an adaptation of the AAPT criteria,¹ as we have reduced the duration of symptoms from 6 to 3 months to reflect the sickle cell disease phenotype better).

Diagnostic Criteria

Reports of ongoing pain present on most days over the past 3 months either in a single location or in multiple locations.

Must display at least 1 sign:

- Palpation of the region of reported pain elicits focal pain or tenderness
- Movement of the region of reported pain elicits focal pain
- Decreased range of motion or weakness in the region of reported pain
- Evidence of skin ulcer in the region of reported pain
- Evidence of hepatobiliary or splenic imaging abnormalities (e.g. splenic infarct, chronic pancreatitis) consistent with the region of reported pain
- Evidence of imaging abnormalities consistent with bone infarction or avascular necrosis in the region of reported pain

There is no other diagnosis that better explains the signs and symptoms.¹

Methodology

Interference in life per acute care visits. Pain and Hurt PedsQL Sickle Cell Disease Module² is utilized (instead of Pain Impact) due to distinctions between mild and severe disease. Patient Reported Outcomes Measurement Information System (PROMIS) T-scores provide thresholds for **pain interference** and **pain behavior** domains, characterizing patients as having no/mild, moderate, or severe pain for both children and adults.³ These PROs are recommended as “core” measurement instruments by Cure Sickle Cell Initiative.⁴ Cutoffs were obtained from pediatric literature and extrapolated to the adult population. AAPT criteria modified after consultation from sickle cell pain expert.

Chronic Pain

Chronic

References

1. Dampier C, Palermo TM, Darbari DS, Hassell K, Smith W, Zempsky W. AAPT Diagnostic Criteria for Chronic Sickle Cell Disease Pain. *J Pain*. May 2017;18(5):490-498. doi:10.1016/j.jpain.2016.12.016
2. Panepinto JA, Torres S, Bendo CB, et al. PedsQL™ sickle cell disease module: feasibility, reliability, and validity. *Pediatr Blood Cancer*. Aug 2013;60(8):1338-44. doi:10.1002/pbc.24491
3. Singh A, Panepinto JA. Clinical meaning of PROMIS pain domains for children with sickle cell disease. *Blood Adv*. Aug 13 2019;3(15):2244-2249. doi:10.1182/bloodadvances.2019000381
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Cognitive Dysfunction

Chronic

Grade 1	IQ evaluation 1-2 SD below mean
Grade 2	IQ evaluation 2-3 SD below mean
Grade 3	IQ evaluation 3 SD or more below mean
Grade 4	N/A
Grade 5	N/A

Definition

Interference or disruption of cognitive processes. This term encompasses a large number of problems and issues associated with intellectual functioning and information processing.¹

Diagnostic Criteria

Validated performance-based tests administered by a trained psychologist or tester including Intelligence measures (specifically the Abbreviated IQ, FSIQ, G, or total cognition composite) from these scales:

- Wechsler Abbreviated Scale of Intelligence, 2nd edition (ages 6+)²
- Wechsler Intelligence Scale for Children, 5th edition (ages 6-16)³
- Wechsler Adult intelligence Scale, 4th Edition (ages 16+)⁴
- Woodcock Johnson Tests of Cognitive Abilities, 4th Edition (ages 2+)⁵
- Kaufman Brief Intelligence Test, 2nd Edition (ages 4+)⁶
- Reynolds Intellectual Assessment Scales (ages 3+)⁷
- Stanford Binet Intelligence Scales, 5th Edition (ages 2+)⁸
- NIH Toolbox, version 3 (ages 7+)⁹

Cognitive Dysfunction

Chronic

Methodology

Intellectual Dysfunction and Executive Dysfunction categories were combined to create a single category; Cognitive Dysfunction. Outcome based on expert guidance from pediatric psychologist.

References

1. Cognitive Dysfunction. Accessed March 25, 2024. http://purl.obolibrary.org/obo/NCIT_C46083
2. Wechsler D. Wechsler Abbreviated Scale of Intelligence-Second Edition. 2nd ed. San Antonio, TX: Pearson; 2011.
3. Gilbert K, Kranzler JH, Benson N. An independent examination of the equivalence of the standard and digital administration formats of the Wechsler Intelligence Scale for Children-5th Edition. J Sch Psychol. Apr 2021;85:113-124. doi:10.1016/j.jsp.2021.01.002
4. Reise SP, Wong E, Block J, et al. Computerized adaptive test strategies for the matrix reasoning subtest of the Wechsler Adult Intelligence Scale, 4th Edition (WAIS-IV). J Int Neuropsychol Soc. Jul 21 2023;1-10. doi:10.1017/S1355617723000401
5. Mather N. Examiner's manual: Woodcock-Johnson III tests of achievement. Itasca, IL: Riverside Publishing. 2001;
6. Kaufman, S A. Kaufman brief intelligence test: KBIT. 1990.
7. Reynolds CR, Kamphaus RW, eds. Handbook of psychological and educational assessment of children: Intelligence, aptitude, and achievement, 2nd ed. The Guilford Press; 2003.
8. Alfonso VC, Flanagan DP. Best practices in the use of the Stanford-Binet Intelligence Scales, Fifth Edition, with preschoolers. In: Alfonso VC, Bracken BA, Nagle RJ, eds. Psychoeducational Assessment of Preschool Children. 2020.
9. Shields RH, Kaat AJ, McKenzie FJ, et al. Validation of the NIH Toolbox Cognitive Battery in intellectual disability. Neurology. 2020;94(12):e1229-e1240. doi:doi:10.1212/WNL.00000000000009131

Elevated TCD Ultrasonography Velocity

Chronic

Grade 1	TAMV 170-184 cm/s (TCD) [i.e. low conditional]
Grade 2	TAMV 185-199 cm/s (TCD) [i.e. high conditional]
Grade 3	TAMV $\geq$ 200 cm/s (TCD) [i.e. abnormal]
Grade 4	N/A
Grade 5	N/A

Definition

A transcranial doppler ultrasound without imaging (TCD) elevation done for evaluation of stroke risk in SCD with abnormal values as shown below.¹

Diagnostic Criteria

TCD ultrasonography measures the velocity of blood flow in the brain. Results are categorized as abnormal, conditional, normal, or inadequate:

- Abnormal TCD: Time averaged mean of the maximum (TAMM) velocity $\geq$ 200 cm/sec in the middle cerebral artery (MCA) or distal internal carotid artery (ICA).
- Conditional TCD: TAMM velocity 170-199 cm/sec in the MCA or distal ICA.
- Normal TCD: TAMM velocity $<$ 170 cm/sec in the MCA or distal ICA.
- Inadequate: On either the right side or left side (or both), no readings can be obtained for the distal ICA, bifurcation, and MCA.

Methodology:

Grading based on velocity of distal internal carotid artery or proximal middle cerebral artery determined by transcranial doppler ultrasound with or without imaging. Because reference ranges for TCDi, as a screening tool for stroke in SCD, have not been established (i.e., the STOP study only investigated non-imaging TCD), we are purposefully not including TCDi to the severity classification grading.

References:

1. Nichols FT, Jones AM, Adams RJ. Stroke prevention in sickle cell disease (STOP) study guidelines for transcranial Doppler testing. J Neuroimaging. Oct 2001;11(4):354-62. doi:10.1111/j.1552-6569.2001.tb00063.x

Posterior Reversible Encephalopathy Syndrome (PRES)

Acute

Grade 1	N/A
Grade 2	N/A
Grade 3	PRES NOT requiring critical cardio-respiratory supportive interventions (e.g., vasoactive/vasopressor infusions, intubation, invasive mechanical ventilation)
Grade 4	PRES requiring critical cardio-respiratory supportive interventions (e.g., vasoactive/vasopressor infusions, intubation, invasive mechanical ventilation)
Grade 5	Death due to complications of PRES

Definition

Posterior reversible encephalopathy syndrome (PRES) is a clinic-radiological syndrome, and its symptoms are a combination of seizures, disturbed vision, altered mental function and headache. It generally presents in the setting of hypertension.¹ PRES is diagnosed with the use of MRI or CT of the head.

Diagnostic Criteria

MRI is the diagnostic gold standard. The predominant affected region is parieto-occipital of both hemispheres. The subcortical white matter is always affected but with frequent involvement of the cortex also. Typical PRES lesions on MRI are thought to represent vasogenic edema.²

Methodology

Grading system begins with Grade 3 and escalates according to use of interventions. PRES is difficult to confirm or grade without an MRI, the diagnostic gold standard,² or CT of head, therefore we did not adopt the CTCAE grading which considers symptoms without a visit to a medical facility.³

Posterior Reversible Encephalopathy Syndrome (PRES)

Acute

References

1. Triplett JD, Kutlubayev MA, Kermode AG, Hardy T. Posterior reversible encephalopathy syndrome (PRES): diagnosis and management. *Practical Neurology*. 2022;22(3):183-189. doi:10.1136/practneurol-2021-003194
2. Roth C, Ferbert A. The posterior reversible encephalopathy syndrome: what's certain, what's new? *Pract Neurol*. Jun 2011;11(3):136-44. doi:10.1136/practneurol-2011-000010
3. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Silent Cerebral Infarct

Chronic

Grade 1	Evidence of one single silent infarct lesion on brain MRI AND NO cognitive deficits
Grade 2	Evidence of > 1 silent infarct lesion on brain MRI AND NO cognitive deficits
Grade 3	Evidence of silent infarct lesion on brain MRI (any number of lesions) AND cognitive deficits
Grade 4	N/A
Grade 5	N/A

Definition

Changes on MRI of brain consistent with infarction without associated history of neurologic symptoms or abnormal neurological exam.¹

Diagnostic Criteria

Must meet all of the following criteria:

- MRI of brain showing increased T2 (or FLAIR) signal in a pattern consistent with ischemia/infarction, ≥ 3 mm in 2 dimensions
- Not explained by other etiologies
- No history of CNS events
- Normal neurologic examination

Note: Cognitive deficit is diagnosed by a performance measure, cognitive testing, or validated patient reported outcome measure for cognitive function.

Silent Cerebral Infarct

Chronic

Methodology

Based on natural history of silent infarct progression and associated clinical complications.^{2,3}

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. DeBaun MR, Armstrong FD, McKinstry RC, Ware RE, Vichinsky E, Kirkham FJ. Silent cerebral infarcts: a review on a prevalent and progressive cause of neurologic injury in sickle cell anemia. Blood. May 17 2012;119(20):4587-96. doi:10.1182/blood-2011-02-272682
3. DeBaun MR, Gordon M, McKinstry RC, et al. Controlled Trial of Transfusions for Silent Cerebral Infarcts in Sickle Cell Anemia. New England Journal of Medicine. 2014;371(8):699-710. doi:10.1056/NEJMoa1401731

Stroke* (Hemorrhagic or Ischemic)

Acute

Grade 1	Incidental radiographic findings only (ischemia, hemorrhage, encephalomalacia, hemosiderosis) AND NO history of neurologic symptoms of stroke; exclude silent cerebral infarct
Grade 2	Neurological symptoms during acute stroke (according to the NIHSS) AND symptoms resolved completely (TIA episode) OR mRS of 0 at discharge
Grade 3	Neurological symptoms during acute stroke (according to the NIHSS) AND symptoms did not resolve completely (remained with at least one neurologic sequela as listed in NIHSS) AND mRS of 1-3 at discharge
Grade 4	Neurological symptoms during acute stroke (according to the NIHSS) AND symptoms did not resolve completely (remained with at least one neurologic sequela as listed in NIHSS) AND mRS of 4-5 at discharge
Grade 5	Death from stroke

Definition

Hemorrhagic: Intracerebral hemorrhage causing neurologic symptoms and signs.

Ischemic: An acute neurological syndrome lasting > 24 hours and resulting from impaired cerebral blood flow without evidence of hemorrhage.¹

Transient Ischemic Attack (TIA): An acute neurological syndrome resulting from impaired cerebral blood flow resolving within 24 hours.

*Overt stroke ONLY: Excludes silent cerebral infarct.

Stroke* (Hemorrhagic or Ischemic)

Acute

Diagnostic Criteria

Hemorrhagic:

- Demonstration of hemorrhage on CT head, MRI of the brain, or autopsy **OR**
- If no imaging is performed, persistently hemorrhagic spinal fluid **AND**
- Rule out hemorrhage caused by traumatic subarachnoid hemorrhage, subdural hematoma, and epidural hematoma

Ischemic:

- MRI or CT scan showing an ischemic CNS event consistent with symptoms and signs **OR**
- Diagnosis by a neurologist based on examination and clinical history with neurologic symptoms/signs lasting > 24 hours

Stroke Scale: Use of National Institute of Health Stroke Scale (NIHSS) at time of presentation and/or Modified Rankin scale score (mRS) at time of discharge

Methodology

Adjusted CTCAE v5 grading system² and NIHSS neurological specifications³ to fit stroke in the setting of SCD. Modified NIHSS 0 score to be Grade 1. mRS utilized as scale (if conducted) upon discharge for disability resulting from stroke.⁴ Score based on age 2-18 (pediatric) or > 18 (adult)⁴. Consideration of ICU in scoring in addition to admissions and interventions. Outcome grading was created with assistance of expert pediatric neurologist consultant.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50
3. NIH Stroke Scale (2003).
4. Bigi S, Fischer U, Wehrli E, et al. Acute ischemic stroke in children versus young adults. Annals of Neurology. 2011;70(2):245-254. doi:https://doi.org/10.1002/ana.22427



Health Outcomes by Organ/System

Eyes & ENT

Hearing Loss (in at least one ear)

Chronic

Grade 1	> 20 dB at > 4 kHz loss Note: > 20 dB at 8 kHz (if 8 kHz is not recorded use 6 kHz)
Grade 2	> 20 dB at ≥ 4 kHz loss Note: > 20 dB at 4 kHz and higher frequencies
Grade 3	> 20 dB at 2 or 3 kHz loss Note: > 20 dB at 2 or 3 kHz and higher frequencies (if 2 kHz is recorded, use 2 kHz, if not, use 3 kHz)
Grade 4	> 40 dB at ≥ 2 kHz, but < 70 dB loss Note: sensorineural hearing loss from 41 to 69 dB at 2 kHz and higher frequencies
Grade 5	≥ 70 dB at ≥ 2 kHz loss Note: sensorineural hearing loss greater or equal to 70 dB at 2 kHz and higher frequencies

Definition

Sensorineural hearing loss: Hearing loss by air and by bone conduction of similar degree, indicating pathology of the cochlea (sensory), nerve (neural), and/or auditory central nervous system (neural).^{1,2}

Diagnostic Criteria

Audiology evaluations and assessments (e.g., audiometry) with normal middle ear function.

Methodology

Grading system based on modifications made to SIOP Boston Ototoxicity Scale and severity of hearing loss from established evaluations.

References

1. Chan Y, Goddard JC. Audiology. KJ Lee's Essential Otolaryngology: Head and Neck Surgery, 12e. McGraw-Hill Education; 2019.
2. Brock PK, Kristin & Freyer, David & Campbell, Kathleen & Steyger, Peter & Blakley, Brian & Chang, Kay & Fligor, Brian & Rajput, Kaukab & Sullivan, Michael & Neuwelt, Edward. Platinum-Induced Ototoxicity in Children: A Consensus Review on Mechanisms, Predisposition, and Protection, Including a New International Society of Pediatric Oncology Boston Ototoxicity Scale. Journal of Clinical Oncology. 2012;30:2408-2417.

Sickle Cell Retinopathy (SCR)

Chronic

Grade 1	Stage 1 or Stage 2 SCR without vision loss
Grade 2	Stage 3 SCR without vision loss OR Stage 1 or Stage 2 SCR with vision loss
Grade 3	Stage 4 or Stage 5 SCR OR any Stage SCR with legal blindness
Grade 4	N/A
Grade 5	N/A

Definition

Proliferative sickle cell retinopathy is a vision-threatening complication of sickle cell disease (SCD). Ischemic events in the retina stimulate angiogenesis, resulting in retinal neovascularization.¹

Non-proliferative sickle cell retinopathy is characterized by arteriolar and capillary occlusion, with accompanying signs such as salmon patch hemorrhages, iridescent spots, and black sunburst lesions.²

Vision loss defined as central visual acuity of > 20/40 and < 20/200 in one or both eyes.³

Legal blindness defined as central visual acuity of 20/200 or worse in both eyes.³

Diagnostic Criteria

Ophthalmologist evaluations and assessments required. Based on staging of sickle cell retinopathy (SCR) as per Goldberg criteria:⁴

Stage 1: Peripheral arterial occlusion

Stage 2: Peripheral arteriovenous anastomoses, representing dilated pre-existing capillaries (hairpin loop)

Sickle Cell Retinopathy (SCR)

Chronic

Stage 3: Neovascular and fibrous proliferation (sea fan) occurring at the posterior border of nonperfusion. A subsequent white sea fan appearance is due to auto-infarction of the neovasculature

Stage 4: Vitreous hemorrhage

Stage 5: Tractional retinal detachment

Methodology

Adopted the Goldberg system which classifies PSR into 5 different stages.⁴ This outcome does not result in death and is a departure from our grading framework. Expert consultancy from adult and pediatric ophthalmologists.

References

1. Akhter M, Latting MW, Scott AW. Management of Proliferative Sickle Cell Retinopathy. EyeNet Magazine. Philadelphia 2018.
2. Campagnoli TR, Krawitz BD, Lin J, et al. Salmon patch-associated vitreous hemorrhage in non-proliferative sickle cell retinopathy masquerading as infectious uveitis. Am J Ophthalmol Case Rep. Mar 2022;25:101329.
3. KY L, FB M. Blindness. StatPearls Publishing; 2024. Updated 2023 Jan 21. <https://www.ncbi.nlm.nih.gov/books/NBK448182/>
4. Goldberg MF. Classification and pathogenesis of proliferative sickle retinopathy. Am J Ophthalmol. Mar 1971;71(3):649-65. doi:10.1016/0002-9394(71)90429-6

Health Outcomes by Organ/System

Gastrointestinal

Cholecystitis/ Cholelithiasis (Gallstones)

Chronic with exacerbation

Grade 1	Cholelithiasis/ choledocholithiasis AND NOT requiring any medical or surgical intervention (e.g., incidental finding)
Grade 2	Cholecystitis or cholelithiasis/ choledocholithiasis AND requiring medical intervention (oral or IV pain medication) AND NOT requiring surgical intervention
Grade 3	Cholecystitis or cholelithiasis/ choledocholithiasis AND requiring surgical intervention in the form of stone retrieval or removal, placement of stent, or cholecystectomy
Grade 4	Cholecystitis or cholelithiasis/ choledocholithiasis AND complicated with pancreatitis OR resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments. ²
Grade 5	Death from cholecystitis or cholelithiasis/ choledocholithiasis complications

Definition

Cholelithiasis: Presence or formation of gallstones in the biliary tract, usually in the gallbladder (cholecystolithiasis) or the common bile duct (choledocholithiasis).¹ In sickle cell disease, cholelithiasis from pigmented gallstones leads to cholecystitis.

Cholecystitis: Inflammation of the gallbladder lining; generally caused by impairment of bile flow, gallstones in the biliary tract, infections, spasm of the gallbladder, or other diseases.¹

Cholecystitis/ Cholelithiasis (Gallstones)

Chronic with exacerbation

Diagnostic Criteria

Cholecystitis:

- Right upper quadrant pain, usually colicky, **AND** one or more of the following:
 - Pericholecystic fluid and gallbladder wall thickening > 4 mm on ultrasonography
 - Non-visualization of the gallbladder by 60 minutes after cholescintigraphy
 - A positive Murphy's sign on physical examination

Cholelithiasis:

- Radiologic (ultrasound) evidence of stones ("hyperechoic with posterior acoustic shadowing") or sludge ("hypoechoic mobile matter within the gallbladder") **OR**
- Pathologic evidence of gallstones or biliary sludge

Methodology

Grading system based on higher levels of care and interventions required. Expert consultancy from pediatric surgeon.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5



Health Outcomes by Organ/System

Genitourinary, Renal, Reproductive

Acute Kidney Injury (AKI)

Acute

Grade 1	sCr 1.5-1.9 times baseline OR increase by ≥ 0.3 mg/dL
Grade 2	sCr 2.0-2.9 times baseline
Grade 3	sCr 3.0 times baseline OR increase in sCr to ≥ 4 mg/dL OR initiation of renal replacement therapy OR (in patients < 18 years) decrease in eGFR to < 35 mL/min per 1.73 m ² AND NO progression to permanent end stage renal disease
Grade 4	sCr 3.0 times baseline OR increase in sCr to ≥ 4 mg/dL OR initiation of renal replacement therapy OR (in patients < 18 years) decrease in eGFR to < 35 mL/min per 1.73 m ² AND progression to end stage renal disease
Grade 5	Death due to AKI

Definition

AKI: An abrupt (within hours) decrease in kidney function, which encompasses both injury (structural damage) and impairment (loss of function).¹

Diagnostic Criteria

- An increase in serum creatinine (sCr) above the upper limit of normal (ULN) for age, present for at least 2 measurements **OR**
- Increase in sCr by 0.3 mg/dL (26.5x mmol/L) within 48 hours **OR**
- Increase in sCr to 1.5 times baseline, which is known or presumed to have occurred within the prior 7 days **OR**
- Urine volume < 0.5 mL/kg/hr for 6 hours²

Kidney Disease Improving Global Outcomes (KDIGO)³ staging utilized.

Acute Kidney Injury (AKI)

Acute

Methodology

Grading system adapted from KDIGO staging with consideration to required therapies and disease progression.

References

1. Makris K, Spanou L. Acute Kidney Injury: Definition, Pathophysiology and Clinical Phenotypes. Clin Biochem Rev. May 2016;37(2):85-98.
2. Nephrology ISO. KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney International Supplements. January 2013 2013;3(1):150. doi:10.1038/kisup.2012.71
3. Group KDIGOKCW. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney Int. Apr 2024;105(4S):S117-S314. doi:10.1016/j.kint.2023.10.018

Acute Papillary Necrosis

Acute

Grade 1	Acute papillary necrosis requiring a visit to medical facility AND NOT requiring treatment AND NOT requiring admission
Grade 2	Acute papillary necrosis requiring a visit to a medical facility AND requiring any treatment (e.g., erythrocyte transfusion, IVF, etc.) AND NOT requiring admission
Grade 3	Acute papillary necrosis requiring admission AND treatment (e.g., interventional procedure to remove RBC's, intubation, mechanical ventilation, erythrocyte transfusions; vaso-pressors) AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 4	Acute papillary necrosis resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to acute papillary necrosis

Definition

Obstruction of the vasa recta of the renal vasculature by sickled erythrocytes. Symptoms include nocturia, dysuria, pyuria, ureteral colic, and back pain.¹

Acute Papillary Necrosis

Acute

Diagnostic Criteria

Acute renal papillary necrosis is a clinical diagnosis that requires a constellation of symptoms, laboratory tests, and diagnostic imaging. Laboratory findings include proteinuria, pyuria, micro or macroscopic hematuria, bacteriuria, and low urine specific gravity. An elevation in serum creatinine may also be present. CT and/or intravenous urography with nephrotomography can support the diagnosis.

Methodology

Grading system based on higher levels of care and interventions required. Acute papillary necrosis can be protracted and last many weeks until it resolves and therefore may require many visits and admissions.

References

1. Renal papillary necrosis. Lancet. Sep 11 1982;2(8298):588-90.
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5

Chronic Kidney Disease (CKD)

Chronic

Grade 1	eGFR 60-89 mL/min/1.73 m ² AND albuminuria < 30 mg/g
Grade 2	eGFR 45-59 mL/min/1.73 m ² AND albuminuria < 30 mg/g OR eGFR ≥ 60 mL/min/1.73 m ² AND albuminuria 30-300 mg/g
Grade 3	eGFR 30-44 mL/min/1.73 m ² AND albuminuria < 30 mg/g OR eGFR 45-59 mL/min/1.73 m ² AND albuminuria 30-300 mg/g OR eGFR 60-89 mL/min/1.73 m ² AND albuminuria > 300 mg/g
Grade 4	eGFR 45-59 mL/min/1.73 m ² AND albuminuria > 300 mg/g OR eGFR 30-44 mL/min/1.73 m ² AND albuminuria > 30 mg/g OR eGFR 15-29 mL/min/1.73 m ²
Grade 5	eGFR < 15 mL/min/1.73 m ² OR Death due to CKD

Chronic Kidney Disease (CKD)

Chronic

Definition

CKD is defined as abnormalities of kidney structure or function, present for > 3 months, with implications for health.¹

Diagnostic Criteria

Either of the following present for > 3 months:

Markers of kidney damage (one or more):

- Albuminuria (ACR $\geq$ 30 mg/24 hours; ACR $\geq$ 30 mg/g [$\geq$ 3 mg/mmol]) at first morning void
- Urine sediment abnormalities
- Electrolyte and other abnormalities due to tubular disorders
- Abnormalities detected by histology
- Structural abnormalities detected by imaging
- History of kidney transplantation

Decreased eGFR

- 60 mL/min/1.73 m² (GFR categories G3a–G5)¹
- GFR 60–89 mL/min/1.73 m² applies to KDIGO Grade 2¹

Methodology

Grading system reflects KDIGO stages of CKD.¹

References

1. Nephrology ISo. KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney International Supplements. January 2013 2013;3(1):150. doi:10.1038/kisup.2012.71

Reproductive Health Complications/Impairments

Female Ovarian Dysfunction

Chronic

Grade 1	Diminished ovarian reserve AND NO evidence of premature ovarian insufficiency
Grade 2	Premature ovarian insufficiency
Grade 3	Infertility
Grade 4	N/A
Grade 5	N/A

Definition

A decrease in ovarian reserve or premature ovarian insufficiency.

Diagnostic Criteria

- Premature ovarian insufficiency: Oligo- or amenorrhea for at least 4 months, in addition to FSH concentration > 25 IU/L on two occasions.²
- Diminished ovarian reserve: Abnormal ovarian reserve test results (antral follicular count (AFC) < 5-7 follicles or AMH below the normal range for age based on the assay used).³
- Infertility: The inability to conceive a pregnancy within 12 months when the female partner is < 35 years old and within 6 months when the female partner is > 35 years old.¹

Note: This condition cannot be graded in prepubertal children.

Methodology

Consultancy with endocrinologist and hematologist specializing in female infertility in hematologic and oncologic diseases.

Reproductive Health Complications/Impairments

Female Ovarian Dysfunction

Chronic

References

1. Webber L, Davies M, Anderson R, et al. ESHRE Guideline: management of women with premature ovarian insufficiency. Hum Reprod. May 2016;31(5):926-37. doi:10.1093/humrep/dew027
2. Ferraretti AP, La Marca A, Fauser BC, Tarlatzis B, Nargund G, Gianaroli L. ESHRE consensus on the definition of 'poor response' to ovarian stimulation for in vitro fertilization: the Bologna criteria. Hum Reprod. Jul 2011;26(7):1616-24. doi:10.1093/humrep/der092
3. Roopa Kanakatti Shankar, Tazim Dowlut-McElroy, Andrew Dauber, Veronica Gomez-Lobo, Clinical Utility of Anti-Mullerian Hormone in Pediatrics, The Journal of Clinical Endocrinology & Metabolism, Volume 107, Issue 2, February 2022, Pages 309-323.

Reproductive Health Complications/Impairments

Male Impairments

Chronic

Grade 1	Intervention not indicated AND infertility despite normal semen parameters AND NO evidence of decreased testicular function (FSH < 7.6 mIU/mL)
Grade 2	Impaired Spermatogenesis (oligospermia or severe oligospermia or Cryptozoospermia) AND evidence of decreased testicular function (FSH level $\geq$ 7.6 mIU/mL) Comment: Might necessitate Intrauterine Insemination (IUI) and/or in vitro fertilization (IVF) + intracytoplasmic sperm injection (ICSI)
Grade 3	Azoospermia AND evidence of decreased testicular function (FSH $\geq$ 7.6 mIU/mL) Comment: Necessitates surgical sperm extraction and IVF + ICSI
Grade 4	N/A
Grade 5	N/A

Definition

Couple infertility may be due to male factors, female factors or a combination of male and female factors therefore parallel evaluation of both partners is always required.¹ See Diagnostic Criteria for defined reproductive health conditions specific to biological or genetic males.

Diagnostic Criteria

Impaired Spermatogenesis: Decreased semen parameters (sperm concentration, motility, morphology, and/or viability) below reference range(s), often with the finding of elevated serum FSH concentration ($>$ 7.6 mIU/mL).

Oligospermia: Semen sample with $<$ 15 million sperm/mL concentration.

Reproductive Health Complications/Impairments

Male Impairments

Chronic

Diagnostic Criteria (continued)

Severe Oligospermia: Semen sample with < 5 million sperm/mL concentration.

Cryptozoospermia: Semen sample with sperm seen only in the centrifuged semen pellet.

Azoospermia: Semen sample with no sperm present.

Methodology

Consultancy with urologist specialized in male infertility.

References

1. Schlegel PN, Sigman M, Collura B, et al. Diagnosis and Treatment of Infertility in Men: AUA/ASRM Guideline Part I. J Urol. Jan 2021;205(1):36-43. doi:10.1097/ju.0000000000001521

Priapism

Acute

Grade 1	Priapism episode with or without the use of medications at home AND NOT requiring a visit to a medical facility
Grade 2	Priapism episode requiring visit to a medical facility AND requiring treatment (e.g., medication) AND NO shunt
Grade 3	Priapism episode requiring a visit to a medical facility AND requiring aspiration/irrigation AND NO shunt
Grade 4	Priapism episode requiring a visit to a medical facility AND requiring shunt surgical intervention
Grade 5	N/A

Definition

A painful, persistent, unwanted erection of the penis, which may recur (stuttering).¹

Diagnostic Criteria

A painful erectile episode event lasting > 30 minutes but diagnosed by a physician or medical practitioner based on patient self-report. During an episode, priapism can be confirmed by physical examination findings of a fully erect penis and complaint of pain in the penis or scrotum.

Methodology

Grading system based on level of care and interventions required.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550

Health Outcomes by Organ/System

Growth & Development

Delayed Puberty

Chronic

Grade 1	N/A
Grade 2	No breast development by age 13 years for females OR testes volume of < 3 cc development by age 14 years for males
Grade 3	No breast development by age 14 years OR no menses by age 16 years or within 4 years of breast development for females OR no increase in testes volume by age 16 years for males OR hormone replacement indicated in either sex
Grade 4	N/A
Grade 5	N/A

Definition

A disorder characterized by unusually late sexual maturity.¹

Diagnostic Criteria

Based on age and breast development for females, and age and testes volume or Tanner Stage 2 development for males.

Methodology

Adapted the CTCAE v5 grading system.¹ Outcome grading created with assistance of expert pediatric endocrinologist consultant.

References

1. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023.

Malnutrition Leading to Stunting (Decreased Height Velocity)

Acute + Chronic

If there is only one data point

Grade 1	N/A
Grade 2	N/A
Grade 3	N/A
Grade 4	Z-score for height-for-age < -3 below mean population
Grade 5	Death

If there are at least 2 data points

Grade 1	N/A
Grade 2	Decline of 1 z-score for height-for-age
Grade 3	Decline of 2 z-score for height-for-age
Grade 4	Decline of 3 z-score for height-for-age
Grade 5	Death

Definition

Pediatric malnutrition (undernutrition) is defined as an imbalance between nutrient requirement and intake, resulting in cumulative deficits of energy, protein, or micronutrients that may negatively affect linear growth, development, and other relevant outcomes. The height-for-age measurement represents the linear growth or stature actually achieved by the child at the age at which the child is measured.

Malnutrition Leading to Stunting (Decreased Height Velocity)

Acute + Chronic

Diagnostic Criteria

Chronic undernutrition or stunting is defined by WHO as having a height-for-age (or length-for-age) that is less than -2 SD (z score) of the median of the NCHS/WHO international reference. Height (i.e., stature) is measured in the standing position. Length-for-age refers to measurements taken in the recumbent position and is recommended for children ≤ 2 years of age.

- For infants aged < 2 years of age, Z-score is used based on WHO reference ranges.²
- For children aged ≥ 2 years of age, Z-score is used based on CDC reference ranges.²

Methodology

Grading based on ability to reach linear growth potential with or without the necessary use of growth supplements. Expert consults from nutritionists.

References

1. World Health Organization. WHO 1995 Expert Committee Report: Physical Status: The Use and Interpretation of Anthropometry. Geneva, Switzerland: World Health Organization; 1995. Technical Report Series 854.
2. Bingham E, Carney L, Sullivan E, et al. Malnutrition (Undernutrition) Clinical Pathway - Inpatient. The Children's Hospital of Philadelphia. Updated March 2023. Accessed April 23, 2024. <https://www.chop.edu/clinical-pathway/malnutrition-undernutrition-clinical-pathway>
3. World Health Organization. WHO 1996 Catalogue of Health Indicators: A Selection of Important Health Indicators Recommended by WHO Programmes. Geneva, Switzerland: World Health Organization; 1996. WHO/HST/SCI/96.8.
4. Mehta NM, Corkins MR, Lyman B, Malone A, Goday PS, Carney LN, Monczka JL, Plogsted SW, Schwenk WF; American Society for Parenteral and Enteral Nutrition Board of Directors. Defining pediatric malnutrition: a paradigm shift toward etiology-related definitions. JPEN J Parenter Enteral Nutr. 2013 Jul;37(4):460-81. doi: 10.1177/0148607113479972. Epub 2013 Mar 25. PMID: 23528324.
5. Consensus Statement of the Academy of Nutrition and Dietetics/American Society for Parenteral and Enteral Nutrition - Becker - 2015 - Nutrition in Clinical Practice - Wiley Online Library
6. Grummer-Strawn LM, Reinold C, Krebs N. Use of World Health Organization and CDC growth charts for children aged 0-59 months in the United States. MMWR Recomm Rep. 2010; 59(RR-9): 1-15.

Underweight

Chronic

Grade 1	N/A
Grade 2	Z-score for weight-for-length or BMI (based on age in diagnostic criteria) 1-1.99 standard deviations below mean population
Grade 3	Z-score for weight-for-length or BMI (based on age in diagnostic criteria) 2-2.99 standard deviations below mean population
Grade 4	Z-score for weight-for-length or BMI (based on age in diagnostic criteria) 3 standard deviations below mean population
Grade 5	Death

Definition

Weight below normal for age.

Diagnostic Criteria

Z-score (for weight for length/height or BMI) greater than or equal to 1 standard deviation (SD) below the mean population reference value (CDC/WHO international reference population).¹

- For infants aged < 2 years of age weight-for-length Z-score is used based on WHO reference ranges.²
- For children aged ≥ 2 years of age, BMI Z-score is used based on CDC reference ranges.²

Methodology

Grading based on ability to reach growth potential with or without the necessary use of growth supplements. Expert consultancy from nutritional and diet professor.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Bingham E, Carney L, Sullivan E, et al. Malnutrition (Undernutrition) Clinical Pathway - Inpatient. The Children's Hospital of Philadelphia. Updated March 2023. Accessed April 23, 2024. <https://www.chop.edu/clinical-pathway/malnutrition-undernutrition-clinical-pathway>



Health Outcomes by Organ/System

Hematologic

Acute Sickle Cell Pain Episode

Acute

Grade 1	Acute pain episode requiring pain treatment at home for pain exacerbation
Grade 2	Acute pain episode requiring a visit to a medical facility (including Day Hospital) for any type of treatment for pain AND NOT requiring admission
Grade 3	Acute pain episode requiring admission AND NO evidence of organ failure or any of the following complications: acute chest syndrome, respiratory distress, DVT, PE, stroke, splenic sequestration, or hyperhaemolysis
Grade 4	Acute pain episode complicated with organ failure OR one of the following complications: acute chest syndrome, respiratory distress, DVT, PE, stroke, splenic sequestration, or hyperhaemolysis OR resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death in the setting of acute pain episode

Definition

A new event of pain or an exacerbation of pain that lasts at least two hours for which there is no explanation other than vaso-occlusion, and which requires therapy with analgesics.¹

Diagnostic Criteria

- Patient self-report **OR**
- Care-giver proxy report for younger children.¹
- **AND** no explanation other than sickle cell disease

Acute Sickle Cell Pain Episode

Acute

Methodology

Grading system based on higher levels of care and interventions required. Definition, Diagnostic Criteria, and Grading reviewed by expert adult hematology consult.³

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5
3. Ataga KI. The challenge of clinical end points in sickle cell disease. Blood. 2023;142(24):2047-2054. doi:10.1182/blood.2023021220

Acute Splenic Sequestration

Acute

Grade 1	Splenic sequestration not requiring any treatment
Grade 2	N/A
Grade 3	Splenic sequestration requiring treatment (i.e. erythrocyte transfusion, IV fluids, etc.) AND NO hemodynamic instability AND NOT requiring splenectomy, fluids, etc.
Grade 4	Splenic sequestration AND hemodynamic instability OR requiring splenectomy
Grade 5	Death due to splenic sequestration

Definition

Rapid intrasplenic trapping of cellular elements of the blood, which causes a precipitous fall in hemoglobin level and is often associated with relative or absolute thrombocytopenia and hypovolemia.¹

Diagnostic Criteria

- Decrease in the hemoglobin or hematocrit of $\geq 20\%$ OR a drop from ≥ 2 g/dL from baseline **AND**
- Evidence of increased erythropoiesis such as a markedly elevated reticulocyte count (exception is when a viral suppression occurs, such as from parvovirus induced transient aplastic crisis) **AND**
- An acutely enlarging spleen (≥ 2 cm from baseline)

Acute Splenic Sequestration

Acute

Methodology

Grading considers existing severity categories (mild: not requiring transfusion, severe: requiring transfusion)¹, modifies the criteria to include the conditions of other treatment (IV fluids), and if a visit to a medical facility was required, and at what level of care.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550

Alloimmunization/ Delayed Hemolytic Transfusion Reaction

Chronic with exacerbation

Grade 1	NOT requiring intervention for immune-mediated hemolysis
Grade 2	Decline in Hb < 20% from baseline AND NOT requiring intervention for immune-mediated hemolysis (including no erythrocyte transfusions, steroids, IVIG, or immunosuppressors)
Grade 3	Decline in hemoglobin ≥ 20% from baseline OR requiring intervention for immune-mediated hemolysis (including erythrocyte transfusions, steroids, IVIG, or immunosuppressors)
Grade 4	Decline in Hb ≥ 20% from baseline AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to alloantibody or autoantibody immune-mediated hemolysis

Definition

Accelerated red blood cell destruction due to antibodies to erythrocyte antigens.¹

Diagnostic Criteria

Drop in Hgb from baseline occurring between 1 and 21 days post transfusion due to hemolysis from alloantibody(ies). The drop in Hgb is associated with immuno-hematology evidence (new RBC antibodies, +/- positive DAT) with evidence of destruction of transfused cells over self RBCs, evidence of prominent hemolysis (increase in LDH, indirect bilirubin, AST), and in some cases relative reticulocytopenia.¹

Alloimmunization/ Delayed Hemolytic Transfusion Reaction

Chronic with exacerbation

Methodology

Grading system based on higher levels of care and interventions required.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5

Chronic Hypersplenism

Chronic

Grade 1	Hypersplenism with isolated thrombocytopenia AND NOT requiring splenectomy
Grade 2	Hypersplenism with 2 cytopenias AND NOT requiring splenectomy
Grade 3	Hypersplenism with all 3 cytopenias OR Hypersplenism leading to splenectomy
Grade 4	N/A
Grade 5	N/A

Definition

Hypersplenism is a condition characterized by splenomegaly, some reduction in the number of circulating blood cells in the presence of normal or hyperactive bone marrow, and the potential for reversal by splenectomy.¹

Diagnostic Criteria

- Splenomegaly (children: palpable ≥ 2 cm below costal margin; adults: palpable) **OR** enlargement of spleen on abdominal imaging **AND**
- One or more cytopenia without evidence of bone marrow insufficiency or effects from hydroxyurea and as defined as:
 - Worsening or new anemia not thought to be related to other etiologies
 - Thrombocytopenia ($< 150,000$ platelets/mm³) not thought to be related to other etiologies
 - Leukopenia or absolute neutrophil count below normal range not thought to be related to other etiologies
 - **AND NO** evidence of acute splenic sequestration

Chronic Hypersplenism

Chronic

Methodology

Grading based on combination of physical exam findings and complications of disease process requiring intervention.

References

1. Hypersplenism. Accessed March 25, 2024. http://purl.obolibrary.org/obo/SCDO_0006905
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. *Ann Intensive Care*. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5

Hepatopathy

Acute

Grade 1	MELD score ≤ 29 AND NOT requiring treatment (e.g., erythrocyte transfusion or drugs for cholestasis)
Grade 2	MELD score ≤ 29 AND requiring treatment (e.g., simple erythrocyte transfusion, cholestatic agents, exchange transfusions) AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ⁵
Grade 3	MELD score 30-39 AND requiring treatment (e.g., simple erythrocyte transfusion, cholestatic agents, or exchange transfusions) AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ⁵
Grade 4	MELD ≥ 40 OR resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ⁵
Grade 5	Death due to liver failure from sickle cell hepatopathy

Definition

Sickle hepatopathy is defined as sickling within hepatic sinusoids leading to vascular stasis and localized hypoxia, which may lead to intrahepatic obstruction of bile formation or flow leading to hyperbilirubinemia.¹ In SCD, this syndrome may occur in the context of hepatic sequestration with the addition of striking hepatic dysfunction and may lead to intrahepatic cholestasis.²

Hepatopathy

Acute

Diagnostic Criteria³

- Abrupt onset of severe right upper quadrant pain **AND/OR**
- Marked increase in direct bilirubin (> 50% of total) compared to baseline **AND**
- Absence of extrahepatic biliary system obstruction **AND/OR**
- Hepatomegaly **AND/OR**
- Elevated liver enzymes **AND/OR**
- Coagulopathy

Model for End-stage Liver Disease (MELD) is a classification used to grade chronic liver disease in adults. The score has prognostic value in terms of three-month mortality.

The components of the MELD score are⁴:

- Serum creatinine (Cr) in mg/dL
 - If dialysis twice in the last week, then creatinine is given a value of 4 mg/dL
- Total bilirubin in mg/dL
- INR

These variables are used to calculate the score with the following formula:

$$\text{MELD} = (0.957 \times \ln [\text{Cr}]) + (0.378 \times \ln [\text{bilirubin}]) + (1.120 \times \ln [\text{INR}]) + 0.643$$

- ($\ln(x) = \log_e(x)$, the natural logarithm of x with base e)

Methodology

Grading system based on higher levels of care and interventions required as well as progressing MELD score for liver failure.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Shah R, Taborda C, Chawla S. Acute and chronic hepatobiliary manifestations of sickle cell disease: A review. World J Gastrointest Pathophysiol. Aug 15 2017;8(3):108-116. doi:10.4291/wjgp.v8.i3.108
3. Gardner K, Suddle A, Kane P, et al. How we treat sickle hepatopathy and liver transplantation in adults. Blood. Apr 10 2014;123(15):2302-7. doi:10.1182/blood-2013-12-542076
4. Townsend Jr CM, Beauchamp RD, Evers BM. Sabiston Textbook of Surgery. 19th ed. Elsevier.
5. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. Doi:10.1186/s13613-024-01283-5

Splenic Infarction

Acute

Grade 1	Incidental finding on radiologic imaging AND NOT requiring treatment
Grade 2	Requiring acute care visit to a medical facility AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 3	Requiring admission for treatment AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 4	Resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ² OR Requiring splenectomy
Grade 5	Death due to acute splenic infarction

Definition

Ischemic necrosis of the spleen because of venous or arterial compromise. In patients with sickle cell disease, this is due to the inability of the rigid sickle red cell to flow within the splenic circulation.¹ Splenic infarction can occur in sickle cell disease secondary to splenomegaly and with inadequate blood supply or occlusion by clot.

Splenic Infarction

Acute

Diagnostic Criteria

- Left upper quadrant pain, which may be referred to the left shoulder **AND**
- Imaging evidence of necrotic or ischemic splenic parenchyma using ultrasound or CT Abdomen **OR**
- Surgical evidence of acute splenic parenchymal necrosis¹

Methodology

Grading system based on higher levels of care and interventions required.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5

Transfusional Iron Overload (Hemochromatosis or Hemosiderosis)

Chronic

Grade 1	LIC 2.5-6.9 mg Fe/100 g dry weight liver OR Serum ferritin 1000-1999 ng/mL (at steady state) irrespective of iron chelation AND MRI T2* $\geq$ 20 msec AND NO evidence of organ dysfunction
Grade 2	LIC 7-14.9 mg Fe/100 g dry weight liver OR Serum ferritin 2000-4999 ng/mL (at steady state) irrespective of iron chelation AND MRI T2* $\geq$ 20 msec AND NO evidence of organ dysfunction
Grade 3	LIC $\geq$ 15 mg Fe/100 g dry weight liver OR serum ferritin 5000-10,000 ng/mL (at steady state) irrespective of iron chelation AND MRI T2* $\geq$ 20 msec AND NO evidence of organ dysfunction
Grade 4	LIC $\geq$ 15 mg Fe/100 g dry weight liver AND serum ferritin $>$ 10,000 ng/mL (at steady state) = irrespective of iron chelation OR MRI T2* $<$ 20 msec OR evidence of organ dysfunction due to iron overload
Grade 5	Death due to iron overload

Transfusional Iron Overload (Hemochromatosis or Hemosiderosis)

Chronic

Definition

Hemochromatosis: A disorder characterized by accumulation of iron in the tissues.¹

Hemosiderosis: An increase in total body iron due to multiple blood transfusions over time.²

Cardiac siderosis: Heart T2* < 20 msec

Diagnostic Criteria

- Liver iron concentration (LIC) > 2.5 mg Fe/g of dry weight liver
- Ferritin > 1000ng/mL at steady state (e.g., no acute fever, infection or acute pain) can be used in the absence of LIC measurement
- Deposition of iron in vital organs resulting in organ dysfunction (i.e. pancreatic dysfunction, liver dysfunction, cardiac siderosis)

Methodology

Grading combines iron overload as defined by the CTCAE v5 grading system¹ in the setting of chronic transfusions indicated for treatment of SCD. Severity stages of I, II, and III (based on liver iron as measured by biopsy)² have been modified to include ASH guideline panel suggestions for screening and management of iron overload (MRI imaging, Ferritin readings, chelation).³

References

1. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50
2. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. *Am J Hematol.* Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
3. Chou ST, Alsawas M, Fasano RM, et al. American Society of Hematology 2020 guidelines for sickle cell disease: transfusion support. *Blood Advances.* 2020;4(2):327-355. doi:10.1182/bloodadvances.2019001143

Transient Aplastic Crisis Secondary To Parvovirus B19 Infection

Acute

Grade 1	Transient aplastic crisis AND NOT requiring treatment (e.g., erythrocyte transfusions, erythropoietin)
Grade 2	Transient aplastic crisis with Hb nadir ≥ 5 g/dL AND requiring treatment (e.g., erythrocyte transfusions, erythropoietin)
Grade 3	Transient aplastic crisis with Hb nadir < 5 g/dL AND requiring treatment (e.g., erythrocyte transfusions, erythropoietin)
Grade 4	Transient aplastic crisis with Hb nadir < 5 g/dL AND requiring treatment (e.g., erythrocyte transfusions, erythropoietin) AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to transient aplastic crisis

Definition

Transient total or partial suppression of erythropoiesis characterized by a decrease in the hemoglobin level and reticulocyte count, or in the number of red blood cell precursors in the bone marrow and positive diagnostic test for parvovirus B19 infection.¹

Transient Aplastic Crisis Secondary To Parvovirus B19 Infection

Acute

Diagnostic Criteria

- Decrease in hemoglobin ≥ 2 g/dL from baseline **AND**
- Reticulocytopenia (absolute reticulocyte count $< 10,000/\text{ul}$) **OR**
- Reticulocyte number disproportionately low with respect to the hemoglobin level¹

Methodology

Grading system based on higher levels of care and interventions required.

References

1. Ballas SK, Lief S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-



Health Outcomes by Organ/System

Infectious Diseases

Fever

Acute

Grade 1	Temperature 38.0 – 38.4 degrees Celsius (100.4 - 101.2 degrees Fahrenheit)
Grade 2	Temperature $\geq$ 38.5 Celsius (101.3 degrees Fahrenheit)
Grade 3	N/A
Grade 4	N/A
Grade 5	N/A

Definition

A disorder characterized by elevation of the body's temperature of 38 Celsius or seeking medication due concern of fever.¹ Ages 0-59 days are excluded.

Diagnostic Criteria

Temperature captured from anywhere. Regardless of presence or absence of central line

OR any patient reported fever

OR any visit/discussion with a medical facility regarding fever

Methodology

Grading system reflects fever in the setting of SCD, which adheres to additional guidelines for initiating and upgrading medical care or interventions.²

References

1. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50
2. Yawn BP, Buchanan GR, Afenyi-Annan AN, et al. Management of sickle cell disease: summary of the 2014 evidence-based report by expert panel members. JAMA. Sep 10 2014;312(10):1033-48. doi:10.1001/jama.2014.10517

Sepsis

Acute

Grade 1	N/A
Grade 2	N/A
Grade 3	Blood culture positive (bacteremia) AND signs or symptoms of organ dysfunction OR treatment indicated
Grade 4	Life-threatening consequences of sepsis (e.g., hemodynamic instability, respiratory failure) AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to sepsis

Definition

Life-threatening organ dysfunction caused by a dysregulated host response to infection.¹

Diagnostic Criteria

- Sepsis Respiratory rate greater than upper limit of normal for age
- Glasgow Coma Scale < 15
- Systolic blood pressure $\leq$ 100 mmHg
- Septic Shock - A need for vasopressor therapy to maintain a mean arterial pressure $\geq$ 65 mmHg and a serum lactate > 2 mM, persisting after adequate fluid resuscitation¹

Sepsis

Acute

Methodology

Grading system based on level of outcome complications, treatment, and therapeutic interventions. Adheres to CTCAE standards.³

References

1. Evans T. Diagnosis and management of sepsis. Clin Med (Lond). Mar 2018;18(2):146-149. doi:10.7861/clinmedicine.18-2-146
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. Ann Intensive Care. Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5
3. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Health Outcomes by Organ/System

Malignancies

Malignant Neoplasms

Acute

Grade 1	Asymptomatic or mild symptoms: Clinical or diagnostic observations only OR low-grade neoplasms where intervention is not indicated (e.g., cervical intraepithelial neoplasia/ cervical dysplasia/ CIN, and teratoma incidentally identified on imaging)
Grade 2	Moderate symptoms: Minimal, local, or noninvasive intervention indicated OR limiting age-appropriate instrumental Activities of Daily Life OR low grade, non-metastatic neoplasms (e.g., cervical carcinoma in-situ, cervical lymph node paraganglioma, basal cell carcinoma, squamous cell carcinoma, parotid carcinoma)
Grade 3	Severe or medically significant but not immediately life-threatening: Hospitalization or prolongation of existing hospitalization indicated OR disabling or limiting self-care Activities of Daily Life OR high-grade neoplasms requiring single treatment therapy (surgery, radiation with or without chemotherapeutic agent) (e.g., breast cancer if in-situ, prostate cancer, bladder carcinoma, thyroid cancer, carcinoid, squamous cell carcinoma cervix, glioma, astrocytoma, gastrointestinal stromal tumor/GIST)

Malignant Neoplasms

Acute

Grade 4	Life-threatening consequences: Urgent intervention indicated OR high-grade neoplasms requiring multimodal therapy OR more than one chemotherapy agent used (e.g., MDS, AML, ALL, Hodgkin lymphoma, non-Hodgkin lymphoma, non in-situ/ invasive breast cancer, osteosarcoma, Ewing sarcoma, primitive neuroectodermal tumor/ PNET, soft tissue sarcoma, renal cancer, anaplastic CNS tumor, glioblastoma, carcinoma of head/neck, liver cancer, lung cancer, mesothelioma, melanoma)
Grade 5	Death due to cancer complications in a person with SCD

Definition

Tumors that have cells that grow uncontrollably and spread locally and/or to distant sites. Malignant tumors are cancerous and spread to distant sites via the bloodstream or the lymphatic system.¹

Diagnostic Criteria

Pathologic diagnosis appropriate to tumor or malignancy type.

Methodology

Adapted from the Grading Rubric for St. Jude Lifetime Cohort² with changes in language to fit the SCOGS format and patient population (i.e., patients with sickle cell disease do not have an increased risk of meningiomas like post-radiation cancer survivors).

References

1. Sobin L, Gospodarowicz, M K., Wittekind, Ch., International Union against Cancer. TNM classification of malignant tumours. Wiley-Blackwell; 2009:310.
2. Hudson MM, Ehrhardt MJ, Bhakta N, et al. Approach for Classification and Severity Grading of Long-term and Late-Onset Health Events among Childhood Cancer Survivors in the St. Jude Lifetime Cohort. Cancer Epidemiol Biomarkers Prev. May 2017;26(5):666-674. doi:10.1158/1055-9965.EPI-16-0812



Health Outcomes by Organ/System

Muscular, Skeletal, Skin

Avascular Necrosis of Joints (AVN)

Chronic

Grade 1	Radiological findings of AVN AND NOT symptomatic
Grade 2	Radiological findings of AVN AND/OR limiting instrumental activities of daily living AND NO intervention
Grade 3	Radiological findings of AVN AND/OR limiting instrumental activities of daily living AND intervention with physical therapy, analgesics, bone remodeling agents OR invasive intervention with steroid injection or core decompression
Grade 4	Radiological findings of AVN AND limiting instrumental activities of daily living AND invasive intervention with total joint replacement
Grade 5	N/A

Definition

Avascular necrosis (AVN) is a condition resulting in dead bone tissue due to an interruption in blood supply, most likely as a result of vaso-occlusion. Bones near a joint, especially the hip, are primarily affected. AVN is also referred to as aseptic necrosis, osteonecrosis, or ischemic necrosis of the bone.¹

Diagnostic Criteria

Radiographic evidence of necrosis and subsequent bone changes. Plain films may be normal early in disease whereas magnetic resonance imaging (MRI) techniques demonstrate early changes and provide more detail on the degree of bony involvement.

Avascular Necrosis of Joints (AVN)

Chronic

Methodology

Adapted CTCAE v5 grading system² to include interventions.

References

1. Ballas SK, Lief S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Leg Ulcer

Chronic with exacerbations

Grade 1	Intact skin with skin induration AND hyperpigmentation associated with pain/itching/burning/ warmth, possibly edema
Grade 2	Partial thickness skin loss AND Less than 8 cm ² AND Minimal exudate AND Periwound intact
Grade 3	Full thickness skin loss involving subcutaneous tissue AND Greater than 8 cm ² < 50% wound bed with necrotic tissue AND Moderate exudate AND Periwound intact
Grade 4	Full thickness skin loss involving damage or necrosis extending to the fascia AND Greater than 8 cm ² ≥ 50% wound bed with necrotic tissue AND Heavy exudate, possibly malodorous AND Periwound erythema, maceration + clinical signs of infection
Grade 5	N/A

Definition

Ulceration of the skin and underlying tissues. Trauma, infection, and severe anemia may predispose patients to ulcer formation.

Diagnostic Criteria

Skin ulceration on physical exam; based on thickness, size, necrosis, periwound, and exudate.¹

Leg Ulcer

Chronic with exacerbations

Methodology

Grading per expert consultancy from adult and pediatric hematologist²

References

1. Monfort JB, Senet P. Leg Ulcers in Sickle-Cell Disease: Treatment Update. Adv Wound Care (New Rochelle). Jun 2020;9(6):348-356. doi:10.1089/wound.2018.0918
2. Minniti CP, Eckman J, Sebastiani P, Steinberg MH, Ballas SK. Leg ulcers in sickle cell disease. Am J Hematol. Oct 2010;85(10):831-3. doi:10.1002/ajh.21838

Osteomyelitis

Acute

Grade 1	Osteomyelitis AND NOT requiring parenteral antibiotics AND NOT requiring invasive treatments (e.g., surgery to drain abscess or bone complications) AND NO multifocal osteomyelitis
Grade 2	Osteomyelitis AND requiring parenteral antibiotics AND NOT requiring invasive treatment (e.g., surgery to drain abscess or bone complications) AND NO multifocal osteomyelitis
Grade 3	Osteomyelitis AND requiring invasive treatment (e.g., surgery to drain abscess or bone complications) AND NO multifocal osteomyelitis
Grade 4	Osteomyelitis AND requiring invasive treatment (e.g., surgery to drain abscess or bone complications) AND multifocal osteomyelitis
Grade 5	Death from complications of osteomyelitis (e.g. sepsis)

Definition

Osteomyelitis is an inflammation of the bone due to infection.¹

Diagnostic Criteria

- Positive cultures from bone obtained by aspiration or biopsy (cultures from a draining tract may not reflect what is in the bone, so biopsies should avoid going through an ulcer) **OR**

Osteomyelitis

Acute

Diagnostic Criteria (continued)

- Positive blood culture accompanied by radiographic and clinical findings consistent with diagnosis¹ **OR**
- Negative blood culture but radiologic and/or histologic findings suggestive of osteomyelitis

Note: this classification does not include chronic osteomyelitis

Methodology

Grading system based on higher levels of care and interventions required.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550

Osteoporosis

Chronic

Grade 1	Adult: Radiologic evidence of osteoporosis or Bone Mineral Density (BMD) t-score -1 to - 2.5 (osteopenia) Pediatric: Radiologic evidence of low BMD with z score of ≤ -2.0 AND NO history of significant fractures for <u>both</u> <u>pediatrics and adults</u>
Grade 2	Adult: BMD t-score < -2.5 OR loss of height < 2 cm or therapy to improve BMD indicated OR limiting instrumental ADL Pediatric: Low BMD (z-score ≤ -2.0) AND significant fracture history (defined as a long bone fracture of the lower extremity, vertebral compression, 2 or more long bone fractures of the upper extremities) for <u>both</u> <u>pediatrics and adults</u> OR therapy to improve BMD indicated for <u>both</u> <u>pediatrics and adults</u>
Grade 3	Adult: Loss of height ≥ 2 cm OR hospitalization indicated OR limiting self-care ADL Pediatric: Limiting self-care ADL
Grade 4	N/A
Grade 5	N/A

Definition

Osteoporosis: Systemic skeletal disease characterized by low bone mass and microarchitectural deterioration of bone tissue with resultant increase in fragility and risk of fracture.¹

Osteoporosis

Chronic

Diagnostic Criteria

Bone mineral density (BMD) by dual-energy X-ray absorptiometry (DEXA) scan.

Methodology

Adopted the CTCAE v5 grading system due to the simplicity² and its severity criteria based on diagnostic criteria of bone mineral density T- and Z-scores.²

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50



Health Outcomes by Organ/System

Other

Non-categorized

Complications

—

Acute Multiorgan Failure

Acute

Grade 1	N/A
Grade 2	N/A
Grade 3	Acute multiorgan failure AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 4	Acute multiorgan failure AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to multiorgan failure

Definition

Acute multiorgan failure syndrome is defined as the acute development of severe dysfunction of at least two of three major organs in the setting of a sickle cell pain episode.¹

Diagnostic Criteria

Acute deterioration in the function of 2 or more organ systems beginning with a > 30% decrease in hemoglobin and platelet count, and involvement of any of the following organ systems utilizing appropriate tests to define the organs involved (examples given in parentheses).

Acute Multiorgan Failure

Acute

Diagnostic Criteria (continued)

Disseminated intravascular coagulopathy (DIC) is also a frequently associated condition:

- Lungs (chest X-ray infiltrate[s], severe hypoxia)
- Liver (increase in ALT, LDH, total and direct bilirubin, alkaline phosphatase)
- Kidney (increased creatinine)
- Central nervous system (altered mental status)
- Heart (ischemia based on characteristic ECG changes or increased troponin)
- Spleen (splenomegaly—ultrasound)
- Coagulopathy (alteration in PT/INR, PTT, and/or fibrinogen)

Methodology

Grading system based on higher levels of care and interventions required.

References

1. Hassell KL, Eckman JR, Lane PA. Acute multiorgan failure syndrome: a potentially catastrophic complication of severe sickle cell pain episodes. *Am J Med.* Feb 1994;96(2):155-62. doi:10.1016/0002-9343(94)90136-8
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. *Ann Intensive Care.* Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5



Health Outcomes by Organ/System

Pregnancy, Puerperium, and Perinatal Conditions

Fetal Growth Restriction

Chronic

Grade 1	N/A
Grade 2	< 10% percentile of weight for gestational age
Grade 3	< 5% percentile of weight for gestational age
Grade 4	< 1% percentile of weight for gestational age
Grade 5	N/A

Definition

A disorder characterized by inhibition of fetal growth resulting in the inability of the fetus to achieve its potential weight.

Diagnostic Criteria

Use of ultrasound during pregnancy to estimate weight of fetus.¹

Note: This grading applies to the mother.

Methodology

Adopted directly from the CTCAE severity grading system.²

References

1. Chia CC, Huang SC. Overview of fetal growth retardation/restriction. Taiwan J Obstet Gynecol. Sep 2014;53(3):435-40. doi:10.1016/j.tjog.2014.01.003
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Pregnancy Loss

Acute

Grade 1	N/A
Grade 2	N/A
Grade 3	N/A
Grade 4	Fetal loss at any gestational age
Grade 5	N/A

Definition

Death in utero.

Diagnostic Criteria

Diagnosis of pregnancy loss is based on self-reporting (delayed menstruation and home pregnancy testing) and clinical testing, including declining serum human chorionic gonadotrophin (hCG) levels, transvaginal ultrasonography and histopathology.¹

Note: This grading applies to the mother.

Methodology

Adopted directly from the CTCAE severity grading system.²

References

1. Dimitriadis E, Menkhorst E, Saito S, Kutteh WH, Brosens JJ. Recurrent pregnancy loss. Nat Rev Dis Primers. Dec 10 2020;6(1):98. doi:10.1038/s41572-020-00228-z
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Premature Delivery

Acute

Grade 1	Delivery of a liveborn infant at > 34 to 37 weeks gestation
Grade 2	Delivery of a liveborn infant at > 28 to 34 weeks gestation
Grade 3	Delivery of a liveborn infant at > 24 to 28 weeks gestation
Grade 4	Delivery of a liveborn infant at 24 weeks of gestation or less
Grade 5	N/A

Definition

A disorder characterized by delivery of a viable infant before the normal end of gestation. Typically, viability is achievable between the twentieth and thirty-seventh week of gestation.

Diagnostic Criteria

Based on gestational age of fetus/infant¹

Note: This grading applies to the mother.

Methodology

Adopted directly from the CTCAE severity grading system.²

References

1. Nour NM. Premature delivery and the millennium development goal. Rev Obstet Gynecol. 2012;5(2):100-5.
2. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50

Health Outcomes by Organ/System

Psychiatric, Psychosocial

Depression

Chronic with exacerbation

Grade 1	Mild depressive symptoms OR PHQ-9 score 5-9 AND treatment not recommended
Grade 2	Moderate depressive symptoms OR PHQ-9 score 10 to 14 AND limiting instrumental activities of daily living AND treatment recommended (e.g., pharmacologic intervention, behavioral therapy)
Grade 3	Severe depressive symptoms OR PHQ-9 score 15-19 AND limiting activities of daily living AND treatment recommended (e.g., pharmacologic intervention, behavioral therapy)
Grade 4	PHQ-9 score 20-27 OR life-threatening consequences (i.e., attempted suicide) OR threats of harm to self or others AND treatment recommended (e.g., pharmacologic intervention, behavioral therapy)
Grade 5	Death due directly to depressive symptoms (e.g., suicide)

Definition

Depression, also known as major depressive disorder or clinical depression, is a common and serious mood disorder. Individuals who suffer from depression experience persistent feelings of sadness, hopelessness, and a loss of interest in activities they once enjoyed. Beyond emotional symptoms, depression can also manifest with physical symptoms such as chronic pain or digestive issues.¹

Depression

Chronic with exacerbation

Diagnostic Criteria

To receive a diagnosis of depression according to the DSM-5 criteria, the following conditions must be met:

1. Duration: Symptoms must be present for at least two weeks.
2. Symptoms: The individual must experience five or more of the following symptoms during the same two-week period, and at least one of these symptoms should be either:
 - Depressed mood: Feeling down most of the day, nearly every day.
 - Loss of interest or pleasure: Markedly diminished interest or pleasure in all or almost all activities.

The additional symptoms include:

- Significant weight loss or gain, or a change in appetite.
- Slowing down of thought and physical movement (observable by others).
- Fatigue or loss of energy.
- Feelings of worthlessness or excessive guilt.
- Diminished ability to think or concentrate, or indecisiveness.
- Recurrent thoughts of death, suicidal ideation, or a suicide attempt.

Methodology

Included because of clinical representation of depression in individuals with SCD.² Modified the CTCAE v5 grading system³ to include interventions and treatment.

References

1. Association AP. Diagnostic and Statistical Manual of Mental Disorders. 5th ed. 2013.
2. Oudin Doglioni D, Chabasseur V, Barbot F, Galactéros F, Gay M-C. Depression in adults with sickle cell disease: a systematic review of the methodological issues in assessing prevalence of depression. BMC Psychology. 2021/04/06 2021;9(1):54. doi:10.1186/s40359-021-00543-4
3. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50



Health Outcomes by Organ/System

Pulmonary

Acute Chest Syndrome (ACS)

Acute

Grade 1	Requiring none of the below: • Supplemental oxygen • Simple or exchange erythrocyte transfusion • Erythropoietin • BiPAP • Any other supportive treatment, other than antibiotics
Grade 2	Requiring FiO₂ > 21% to < 50% OR simple erythrocyte transfusion AND none of the below: • BiPAP • High flow O₂ • Exchange transfusion
Grade 3	Requiring ≥ 50% FiO₂ OR BiPAP OR high flow O₂ OR exchange transfusion
Grade 4	Requiring ≥ 50% FiO₂ OR BiPAP OR high flow O₂ OR exchange transfusion AND critical cardio-respiratory supportive interventions (e.g., vasoactive/vasopressor infusions, intubation, invasive mechanical ventilation)
Grade 5	Death from ACS complications

Acute Chest Syndrome (ACS)

Acute

Definition

ACS is an acute illness characterized by fever and/or respiratory symptoms, accompanied by a new LOBAR pulmonary infiltrate (not atelectasis) on a chest imaging.¹

Diagnostic Criteria

Radiographic evidence of consolidation. A new segmental radiographic pulmonary infiltrate (not requiring preceding radiograph or physical exam, but if either performed, current findings must be new) involving at least one complete segment **AND** at least one of the following::

- Temperature $\geq 38.5^{\circ}\text{C}$
- $> 2\%$ decrease in SpO₂ (HbO₂ saturation) from a documented steady-state value on room air (FiO₂ = 0.21%)
- PaO₂ < 60 mmHg
- Tachypnea (per age-adjusted normal)
- Intercostal retractions, nasal flaring, or use of accessory muscles of respiration
- Thoracic pain
- Cough
- Wheezing
- Rales

Methodology

Grading system was created based on evidence within sickle cell disease literature.² Severity is according to the level of care and complexity of treatment/s required.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Hankins J, Jeng M, Harris S, Li CS, Liu T, Wang W. Chronic transfusion therapy for children with sickle cell disease and recurrent acute chest syndrome. J Pediatr Hematol Oncol. Mar 2005;27(3):158-61. doi:10.1097/01.mph.0000157789.73706.53

Asthma Exacerbation

Acute

Grade 1	Mild symptoms AND NO medical intervention indicated
Grade 2	Symptomatic AND medical intervention sought (but NO supplemental oxygen) OR limiting instrumental activities of daily living (ADL) OR intermittent asthma requiring short-acting beta-agonists as needed AND NOT status asthmaticus
Grade 3	Limiting self-care ADL OR supplemental oxygen indicated OR status asthmaticus
Grade 4	Life-threatening respiratory or hemodynamic compromise (i.e., requiring intubation, pressors, invasive or non-invasive ventilation)
Grade 5	Death due to asthma complications

Definition

Acute or subacute episodes of progressively worsening shortness of breath, cough, wheezing, and chest tightness—or some combination of these symptoms with underlying diagnosis of asthma.¹

Diagnostic Criteria

- Episodic symptoms of airflow obstruction or airway hyperresponsiveness are present
- Airflow obstruction is at least partially reversible
- Alternative diagnoses are excluded
- If spirometry is available, reversibility is determined either by an increase in FEV1 of ≥ 12 percent from baseline or by an increase ≥ 10 percent of predicted FEV1 after inhalation of a short-acting bronchodilator

Asthma Exacerbation

Acute

Diagnostic Criteria (continued)

- Additional studies as necessary to exclude alternate diagnoses¹
- **Improvement with bronchodilators**

Methodology

The classification and severity guidelines were adapted from the SJLIFE outpatient focused grading approach.²

References

1. National Asthma E, Prevention P. Expert Panel Report 3 (EPR-3): Guidelines for the Diagnosis and Management of Asthma-Summary Report 2007. J Allergy Clin Immunol. Nov 2007;120(5 Suppl):S94-138. doi:10.1016/j.jaci.2007.09.043
2. Hudson MM, Ehrhardt MJ, Bhakta N, et al. Approach for Classification and Severity Grading of Long-term and Late-Onset Health Events among Childhood Cancer Survivors in the St. Jude Lifetime Cohort. Cancer Epidemiol Biomarkers Prev. May 2017;26(5):666-674. doi:10.1158/1055-9965.EPI-16-0812

Chronic Restrictive Lung Physiology

Chronic

Grade 1	TLC > 70% and TLC ≤ 80% %; mild restrictive disease
Grade 2	TLC > 60% and TLC ≤ 70%; moderate restrictive disease
Grade 3	TLC ≥ 50% and TLC ≤ 60%; moderately severe restrictive disease
Grade 4	TLC < 50% predicted; severe restrictive lung disease
Grade 5	Death due to restrictive lung disease

Definition

Chronic shortness of breath, cough, wheezing, and chest tightness with known underlying chronic restrictive lung disease defined as a group of heterogenous pulmonary disorders characterized by a reduced distensibility of the lungs, compromising lung expansion, and, in turn, reduced lung volumes, particularly with reduced total lung capacity (TLC).¹

Diagnostic Criteria

Functional characteristics of a restrictive pattern in pulmonary function tests (PFTs) which include decreased total lung capacity (TLC) and forced vital capacity (FVC). According to the American Thoracic Society, the TLC predicted (adjusted for gender, age, height) value can be used for assessing the severity of the restrictive condition.² For patients with sickle cell disease, PFTs should be taken at baseline and not in the setting of acute illness.

Methodology

Grading system created with assistance of expert adult pulmonologist and based on guidance from the American Thoracic Society.

Chronic Restrictive Lung Physiology

Chronic

References

1. Hariri LP, Smith ML, Mino-Kenudson M, et al. Pulmonary Pathology Society Perspective on the 2018 American Thoracic Society, European Respiratory Society, Japanese Respiratory Society, and Latin American Thoracic Society Idiopathic Pulmonary Fibrosis Clinical Practice Guidelines. Ann Am Thorac Soc. May 2020;17(5):550-554. doi:10.1513/AnnalsATS.201910-801PS
2. Aggarwal AN, Agarwal R. The new ATS/ERS guidelines for assessing the spirometric severity of restrictive lung disease differ from previous standards. Respirology. Sep 2007;12(5):759-62. doi:10.1111/j.1440-1843.2007.01117.x

Pulmonary Embolism (PE)

Acute

Grade 1	An incidental finding of PE or diagnosis of PE AND NO symptoms
Grade 2	Finding of PE on imaging AND symptoms AND NO cardiac function compromise AND NO hemodynamic instability
Grade 3	Finding of PE requiring admission AND symptoms AND cardiac function compromise, including evidence of right heart strain by ECHO, EKG, or elevated BNP (from baseline) AND NO hemodynamic instability AND NOT resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 4	Finding of PE requiring ICU admission AND symptoms AND cardiac function compromise, including evidence of right heart strain by ECHO, EKG, or elevated BNP (from baseline) OR hemodynamic instability AND resulting in a life-threatening complication where the need for life-supporting treatment indicated, including use of mechanical ventilation, vasopressors, renal replacement therapy, or other life-supporting treatments ²
Grade 5	Death due to PE

Definition

Embolism from the systemic circulation to the pulmonary vasculature.

Alternately, thrombi may form in situ. Pulmonary embolism may cause isolated dyspnea, pleuritic chest pain, hemoptysis, or circulatory collapse.¹

Pulmonary Embolism (PE)

Acute

Diagnostic Criteria

ANY of the following:

- Pulmonary angiography demonstrating an intraluminal filling defect or abrupt vessel cut-off.
- High probability ventilation/perfusion (V/Q) lung scan showing:
 - Two or more large segmental perfusion defects (> 75% of a segment) without corresponding ventilation or radiographic abnormalities.
 - One large segmental perfusion defect and 2 or more moderate segmental perfusion defects (25%-75% of a segment) without corresponding ventilation or radiographic abnormalities.
 - Four or more moderate segmental perfusion defects without corresponding ventilation or radiographic abnormalities.
- CT scan (helical or spiral CT) with contrast medium demonstrating intraluminal defect(s) that partially or completely occlude a pulmonary artery or result in an abrupt vessel cut-off.
- Gadolinium-enhanced pulmonary magnetic resonance angiography demonstrating intraluminal defect(s) that partially or completely occlude a pulmonary artery or an abrupt vessel cut-off.

Methodology

Grading system based on higher levels of care and interventions required, with consideration to statements from the American Heart Association.³ Outcomes created with assistance of expert adult pulmonologist.

References

1. Ballas SK, Lieff S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. *Am J Hematol.* Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
2. Levy M, Naudin J, Geslain G, et al. Factors associated with adverse outcome among children with sickle cell disease admitted to the pediatric intensive care unit: an observational cohort. *Ann Intensive Care.* Apr 10 2024;14(1):55. doi:10.1186/s13613-024-01283-5
3. Giri J, Sista AK, Weinberg I, et al. Interventional Therapies for Acute Pulmonary Embolism: Current Status and Principles for the Development of Novel Evidence: A Scientific Statement From the American Heart Association. *Circulation.* Nov 12 2019;140(20):e774-e801. doi:10.1161/cir.0000000000000707

Pulmonary Hypertension

Chronic

Grade 1	Right heart catheterization mPAP 20-25 mmHg AND NYHA Class I-II dyspnea AND NO evidence of right sided congestive heart failure or syncope
Grade 2	Right heart catheterization mPAP 25-35 mmHg AND NYHA Class II-III dyspnea and/or persistent cough AND NO evidence of right sided congestive heart failure or syncope
Grade 3	Right heart catheterization mPAP > 35 mmHg AND NYHA Class III dyspnea and/or hypoxia requiring nocturnal or constant oxygen AND NO evidence of right sided congestive heart failure on physical exam
Grade 4	Right heart catheterization mPAP > 35 mmHg AND NYHA Class IV dyspnea (syncope and/or dyspnea at rest) AND right sided congestive heart failure on physical exam AND severely reduced cardiac output/cardiac index or respiratory failure requiring ventilatory support
Grade 5	Death in the setting of pulmonary hypertension

Definition

A disorder characterized by an increase in pressure within the pulmonary circulation due to lung or heart disorder.¹

Diagnostic Criteria

Direct measurement of pulmonary artery pressure by right-sided cardiac catheterization is the gold standard test for the diagnosis of pulmonary arterial hypertension (PASP > 25 mmHg at rest or > 30 mmHg with exercise).²

Pulmonary Hypertension

Chronic

Methodology

Grading system based on the combination of the New York Heart Association (NYHA) classes I-IV of dyspnea^{3,4} and the requirements of medical intervention per the CTCAE v5.¹ Expert consultancy from adult pulmonologist was used.

- NYHA Class I dyspnea: no evidence of right sided congestive heart failure or syncope, right heart cath – mPAP 20-25 mmHg, normal cardiac output/cardiac index, normal right atrial pressure
- NYHA Class II dyspnea: no evidence of right sided congestive heart failure or syncope. right heart cath – mPAP 25-35 mmHg, normal cardiac output/cardiac index, normal right atrial pressure
- NYHA Class III dyspnea: no evidence of right sided congestive heart failure or syncope, right heart cath - mPAP > 35 mmHg, reduced cardiac output/cardiac index, right atrial pressure > 10 mmHg
- NYHA Class IV dyspnea: (syncope and/or dyspnea at rest), right sided congestive heart failure, mPAP > 35 mmHg, severely reduced cardiac output/cardiac index, right atrial pressure > 14 mmHg, shock, respiratory failure requiring ventilatory support⁴

Note: If patient fits criteria for different grades, the highest grade will be applied. For instance, if patient fits Grade 2 criteria based on mPAP, but Grade 3 based on NYHA and echocardiogram criteria, this patient will be classified as Grade 3 pulmonary hypertension.

References

1. Services USDoHaH. Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0. Accessed February 24, 2023. ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm#ctc_50
2. Ballas SK, Lief S, Benjamin LJ, et al. Definitions of the phenotypic manifestations of sickle cell disease. Am J Hematol. Jan 2010;85(1):6-13. doi:10.1002/ajh.21550
3. Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J. Oct 11 2022;43(38):3618-3731. doi:10.1093/eurheartj/ehac237
4. Commission TJ. New York Heart Association (NYHA) Classification. Updated January 10, 2023. Accessed 4/10/23, 2023. <https://manual.jointcommission.org/releases/TJC2016A/DataElem0439.html>.

Sleep Apnea (Obstructive or Central)

Chronic

Adults (age ≥ 18)

Grade 1	Polysomnography AHI 5-15 AND NO treatment indicated
Grade 2	Polysomnography AHI 5-15 AND treatment indicated (i.e. CPAP, tonsillectomy)
Grade 3	Polysomnography AHI > 15-30 with or without treatment (i.e. CPAP, tonsillectomy) indicated
Grade 4	Polysomnography AHI > 30 with or without treatment indicated (i.e. CPAP, tonsillectomy)
Grade 5	N/A

Children (age < 18 yrs)

Grade 1	AHI 0 – 3 AND HbO2 sat < 90% for > 3 min overnight OR AHI 3.1 – 7.9 AND NO HbO2 sat < 90% for > 3 min overnight
Grade 2	AHI 3.1 – 7.9 AND HbO2 sat < 90% for > 3 min overnight OR AHI 8.0 – 15.0 AND NO HbO2 sat < 90% for > 3 min overnight
Grade 3	AHI 8.0 – 15.0 AND HbO2 sat < 90% for > 3 min overnight OR AHI 15.1 – 24.9 AND NO HbO2 sat < 90% for > 3 min overnight

Sleep Apnea (Obstructive or Central)

Chronic

Children (age ≤ 18 yrs) *continued*

Grade 4	AHI 15.1 – 24.9 AND HbO2 sat < 90% for > 3 min overnight OR AHI ≥ 25
Grade 5	N/A

Definition

Obstructive Sleep Apnea: A condition characterized by obstruction of the airway and by pauses in breathing during sleep occurring many times during the night. Obstructive sleep apnea is related to a relaxation of muscle tone (which normally occurs during sleep) leading to partial collapse of the soft tissues in the airway with resultant obstruction of the air flow.¹

Central Sleep Apnea: A sleep apnea that is characterized by a malfunction of the basic neurological controls for breathing rate and the failure to give the signal to inhale, causing the individual to miss one or more cycles of breathing.²

Diagnostic Criteria

- Polysomnography measuring apnea-hypopnea index (AHI).
- Apnea-Hypopnea Index measures sleep apnea severity.
- The AHI is the sum of the number of apneas (pauses in breathing) plus the number of hypopneas (periods of shallow breathing) that occur, on average, each hour

Methodology

Grading according to apnea-hypopnea index recommendations³ and severity of resulting conditions.

Sleep Apnea (Obstructive or Central)

Chronic

References

1. Obstructive Sleep Apnea. Accessed April 15, 2024. http://purl.obolibrary.org/obo/HP_0002870
2. Central Sleep Apnea. Accessed April 15, 2024. http://purl.obolibrary.org/obo/DOID_9220
3. Force AAoSMT. Sleep-related breathing disorders in adults: recommendations for syndrome definition and measurement techniques in clinical research. The Report of an American Academy of Sleep Medicine Task Force. *Sleep*. Aug 01 1999;22(5):667-89.

Grading Frequency Guide

Grading Frequency Guide

Outcome Classification	ACUTE	CHRONIC	CHRONIC WITH EXACERBATIONS
Acute chest syndrome (ACS)	X		
Acute kidney injury (AKI)	X		
Acute multiorgan failure	X		
Acute papillary necrosis	X		
Acute sickle cell pain episode	X		
Acute splenic sequestration	X		
Alloimmunization/Delayed Hemolytic Transfusion reaction			X
Arrhythmia			X
Asthma exacerbation	X		
Avascular necrosis of joints (AVN)		X	
Cerebral vasculopathy		X	
Cholecystitis/ Cholelithiasis (gallstones)			X
Chronic hypersplenism		X	
Chronic kidney disease (CKD)		X	
Chronic pain		X	
Chronic restrictive lung physiology		X	
Cognitive dysfunction		X	
Deep Vein Thrombosis (DVT)	X		
Delayed puberty		X	
Depression			X
Diastolic dysfunction		X	

Grading Frequency Guide

Outcome Classification	ACUTE	CHRONIC	CHRONIC WITH EXACERBATIONS
Elevated TCD ultrasonography velocity		X	
Fetal growth restriction <i>Note: Grade each ultrasound; each pregnancy is a chronic episode</i>		X	
Fever	X		
Hearing loss (in at least one ear) <i>Note: Grade each ear; considered separate organs</i>		X	
Heart failure exacerbation	X		
Hepatopathy	X		
Leg ulcer <i>Note: Each ulcer graded as its own episode</i>			X
Malnutrition leading to stunting (decreased height velocity) <i>Note: < 6 month duration is Acute, ≥ 6 month duration is Chronic</i>	X	X	
Malignant neoplasms	X		
Myocardial infarction	X		
Osteomyelitis	X		
Osteoporosis		X	
Posterior reversible encephalopathy syndrome (PRES)	X		
Pregnancy loss	X		
Premature delivery	X		
Priapism	X		
Pulmonary embolism	X		

Grading Frequency Guide

Outcome Classification	ACUTE	CHRONIC	CHRONIC WITH EXACERBATIONS
Pulmonary hypertension		X	
Reproductive health complications: female ovarian dysfunction		X	
Reproductive health complications: male impairments		X	
Sepsis	X		
Sickle retinopathy (SCR)		X	
Silent cerebral infarct		X	
Sleep apnea (obstructive or central)		X	
Splenic infarction	X		
Stroke (hemorrhagic or ischemic)	X		
Systemic arterial hypertension		X	
Systolic dysfunction		X	
Transfusional iron overload		X	
Transient aplastic crisis secondary to parvovirus B19 infection	X		
TRV elevation on echocardiogram		X	
Underweight		X	

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SCOGS Investigators

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Operational Team

SCOGS was operationalized by a multidisciplinary team of hematologists, oncologists, clinicians, researchers, biostatisticians, data scientists, and program support specialists based at St. Jude Children's Research Hospital.

This team developed the original concept, drafted a system of classifications, built eDelphi rounds, analyzed, synthesized, and shared results with content matter experts, incorporated collaborative modifications between rounds, implemented severity scoring in-house, and facilitated communication across institutions to build and share the development of SCOGS. Thank you to these individuals for their contributions.

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John Strouse, MD, PhD

Duke University

Ravi Talati, DO

Cleveland Clinic

Clifford Takemoto, MD

St. Jude Children's Research Hospital

Madhav Vissa, MD

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Winfred Wang, MD

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Specialty Consultants

Acute Sickle Cell Pain

Kenneth Ataga, MD

*University of Tennessee Health Science
Center for Sickle Cell Disease*

Cholecystitis / Cholelithiasis (Gallstones)

Abdelhafeez Abdelhafeez, MD

*St. Jude Children's Research Hospital /
University of Rochester Medical Center*

Chronic Kidney Disease, Acute Kidney Injury

Jeff Lebensburger, DO

University of Alabama at Birmingham

Chronic Pain

Deepika Darbari, MD

*Children's National Hospital / GW School of
Medicine & Health Sciences*

Delayed Puberty

Angela Delaney, MD

St. Jude Children's Research Hospital

Diastolic Dysfunction

Fredrick L. Ruberg, MD

Boston Medical Center, Boston University

Ankit A. Desai, MD

Indiana University School of Medicine

Elevated TRV

Jason Johnson, MD, MHS

*Le Bonheur, University of Tennessee
Health Science Center*

Fever

Elsbeth Bittle, MD, MSc

St. Jude Children's Research Hospital

Sheena Mukkada, MD, MPH

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Specialty Consultants (continued)

Hearing Loss

Johnnie Bass, AuD, PhD

St. Jude Children's Research Hospital

Celine Richard, MD, PhD

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St. Jude Children's Research Hospital*

Malnutrition

Maria R. Mascarenas, MBBS

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Babette S. Zemel, PhD

Children's Hospital of Philadelphia

Malnutrition, Underweight

Michell Fullmer, RD

Nemours Children's Health

Myocardial Infarction

Ankit A. Desai, MD

Indiana University School of Medicine

Reproductive Health Complications/Impairments (female)

Kari Bjornard, MD

*Riley Children's Health, Indiana University,
St. Jude Children's Research Hospital*

Robert E. Brannigan, MD

*Northwestern University, Feinberg School of
Medicine, Northwestern Medicine*

Lydia H. Pecker, MD

Johns Hopkins Medicine

Reproductive Health

Complications/Impairments (male)

Mary Elaine Killian, MD

*UT Le Bonheur, University of Tennessee
Health Science Center*

Sickle Retinopathy (Proliferative or Non-proliferative)

Mary Ellen Hoehn, MD

*Methodist / Le Bonheur / University of
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St. Jude Children's Research Hospital*

Benjamin King, MD

University of Tennessee Health Science Center

Stroke (Hemorrhagic or Ischemic)

Lori Jordan, MD, PhD

Vanderbilt University Medical Center

Operational Team

Emory University School of Medicine

Tarun Aurora, MD, MSCI

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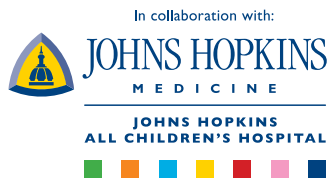
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Sickle Cell Center
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